NarrativeWiki: Probing the Trade-offs of Decomposition in Graph-Built Character Wikis at Reader-Selected Narrative Cutoffs

Anonymous ACL submission

Abstract

A character wiki that reveals only what a reader has reached invites decomposing the novel into a graph of claims filtered by volume. We instrument this design as NARRATIVEWIKI and test it against six systems at every cutoff of a five-volume series (111 gold facts; LLM-finalized labels; two human reviewers relabeled 554 original-text items). A renamed and paraphrased copy (the treated text) tests recall: closed-book recall falls from 0.683 to 0.003, while systems given text lose at most 0.070. Here the design trades context for accumulation. Rendered without a length budget, NARRATIVEWIKI's pages grow with the story: on treated text it best covers the most-documented characters (recall 0.929; long context 0.794), loses one fact across cutoffs per text (others: 4–24), and publishes 3.8 times as many distinct assertions. But claims lose their context: NARRATIVEWIKI inverts a relation, turns a prediction into a marriage, groups entities unlike prefix-built indexes (0.220 of first-cutoff pages, original text) and misplaces true statements. Prompting with only the text read discloses no assessed later fact; an instruction to withhold later volumes discloses 17/88 later facts on treated text (original: 8/88). Code, gold facts and labels: `https://anonymous.4open.science/r/NarrativeWiki-Anon-341E/`.

1 Introduction

A reader consulting a character wiki midway through a series may encounter a later identity, marriage or death while seeking a reminder of events already read. A reader who has finished *Anne of Avonlea*, the second book of L. M. Montgomery's Anne series, may learn from Anne Shirley's whole-series page about her marriage to Gilbert Blythe in book four (Figure 1).[1] A useful reference must respect the reader's progress: at cutoff t, every statement should be established by the first t volumes.

The natural engineering answer is structural. Decompose the text into a graph of claims, date each claim by the volume of its evidence, and generate each page from the claims dated no later than the cutoff, in the manner of graph-based retrieval-augmented generation (Edge et al., 2024; Guo et al., 2025). The boundary is then enforced by data, not by an instruction to the generator, and one build serves every cutoff. This paper asks what that decomposition gains and what it costs.

We instrument this design rather than advocate for it. NARRATIVEWIKI is an *analytical probe*: it adds verbatim-evidence gates, a claim verifier and logged model calls, which lets us trace individual errors to the stage that produced them, and it is placed among systems that differ in what the generator can access and in how pages are written (Figure 6).

Such a comparison is interpretable only if scores reflect reading: given only a title and a character name, the generator recovers 0.683 of the eligible gold facts. We therefore pair every measurement with a *renamed-text control*: a *treated* copy of the series renames characters and places and paraphrases nearly every paragraph, keeping paragraph identifiers so that gold facts and citations stay aligned. Closed-book recall falls to 0.003, while systems given text largely keep theirs (Section 6.1).

Headline means rank NARRATIVEWIKI below prompting with the text read so far (Table 1). The probe asks instead *where* decomposition gains and *where* a volume filter fails, through five research questions, each answered in the subsection of Section 6 with its number (Table 2):

- **RQ1 Recognition vs. reading.** How much apparent recall comes from knowing the work? Closed-book recall falls from 0.683 to 0.003 under the control.

[1] Examples in this paper reveal plot points of the series.

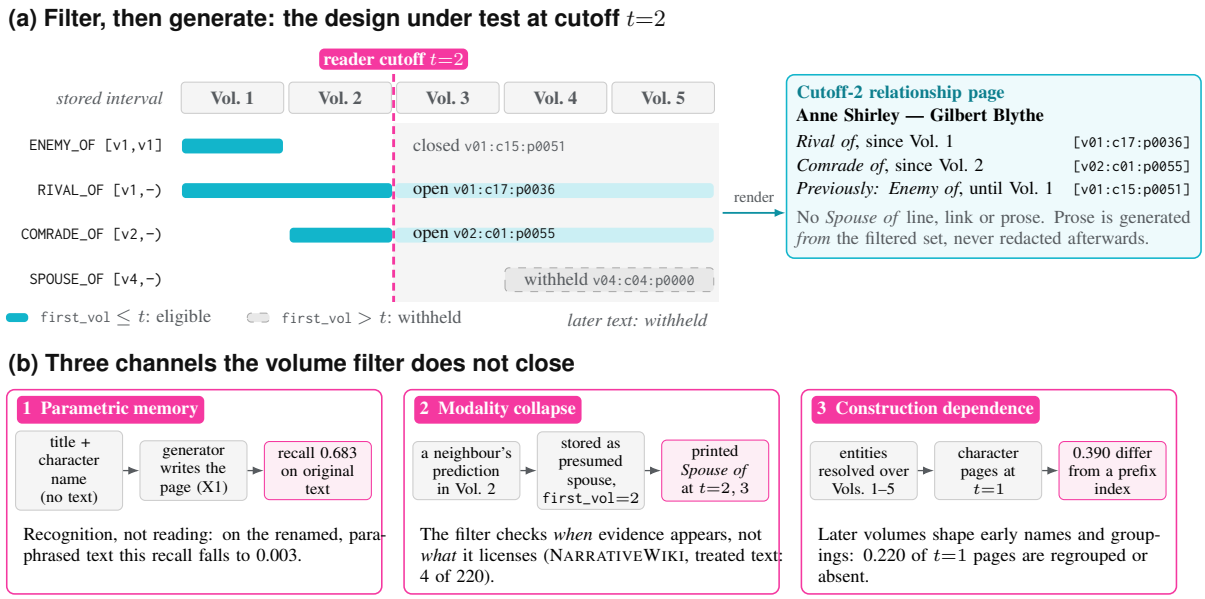


Figure 1: Overview. (a) The filter-before-generate design that NarrativeWiki implements and this paper probes, on the four stored Anne–Gilbert intervals: at cutoff 2 only intervals whose first evidence volume is at most 2 reach the page, so the volume-4 marriage is withheld. (b) The three channels through which later information still reaches a reader, each with its measured size (Sections 6.1, 6.2 and 6.5).

- **RQ2 Cutoff enforcement.** Does a volume filter prevent disclosure? A neighbour's prediction becomes a marriage two volumes early (*modality collapse*).
- **RQ3 Accumulation.** What do accumulated pages gain? The best recall on the most-documented characters and one fact lost across cutoffs per text.
- **RQ4 Context.** Do losses arise in the graph or the page? True statements lose support where the graph places them (*context collapse*).
- **RQ5 Construction scope.** Do whole-series entities shape early pages? 0.390 of pages at $t = 1$ differ from a prefix-built index.

Prior work bounds dialogue or answers in time (Ahn et al., 2024; Huang et al., 2025). We study a linked reference artifact at every cutoff:

- **A probe study of structural decomposition** under a reading cutoff, separating its gains from failures that a volume filter cannot detect (Sections 4, 6).
- **A paired renamed-text control** as evaluation methodology: seven systems rebuilt on aligned treated text with unchanged code, prompts and gold facts, and recognition measured separately from recall (Section 6.1).
- **A task, threat model and artifact-level protocol** for cutoff-bounded reference generation, with 111 gold facts dated by reveal volume (Sections 3, 5; Appendix C).

2 Related Work

Spoilers, characters and temporal bounds. Spoiler detection labels review sentences without conditioning on a reader's position (Wan et al., 2019); Jones et al. (2018) serve fan-wiki revisions archived before an episode aired, and e-readers offer position-bounded answers (Amazon Staff, 2025) without a published evaluation. CHIRON builds validated character sheets (Gurung and Lapata, 2024), BookWorm pairs books with character descriptions (Papoudakis et al., 2024), and EvolvTrip graphs characters' mental states over time (Yang et al., 2025). TimeChara measures point-in-time hallucination in role-play (Ahn et al., 2024); Living the Novel gates conversation to the narrative present (Huang et al., 2025), the filter-before-generate principle we adopt; NS-ST-GraphRAG answers from temporally valid subgraphs (Lin, 2026); temporal knowledge graphs attach validity intervals to facts (Dasgupta et al., 2018), ours indexed by reading volume rather than story time. These works bound answers or dialogue in time; none evaluates a complete linked reference artifact at every cutoff (Table 3). LitBank annotates identities using the whole text, including revelations unavailable at an earlier mention (Bamman

et al., 2020).

Grounded article and wiki generation. STORM outlines Wikipedia-like articles (Shao et al., 2024); GraphRAG summarizes a corpus entity graph (Edge et al., 2024); LightRAG, a baseline here, combines graph structure with lightweight retrieval (Guo et al., 2025); SearchWiki builds linked wikis for agents (Singh et al., 2026); book-length benchmarks evaluate question answering and summarization (Wang et al., 2025; Chang et al., 2024). Our reference also depends on the reader's position.

Factuality, attribution and contamination. We evaluate atomic support (Min et al., 2023), attribution to identified sources (Rashkin et al., 2023) and citation quality (Gao et al., 2023) across the whole artifact and in page context. Automatic raters of book-length text can disagree with knowledgeable readers (Kim et al., 2024); on STORYSUMM, MiniCheck's Flan-T5 checker (Tang et al., 2024) scored below chance (Subbiah et al., 2024), so we use evidence-reading annotators. Published novels may be known to the generator (Chang et al., 2023; Cheng et al., 2025); renaming entities is an established control (Gupta, 2025; Zhang et al., 2024), and rewriting can damage the narrative it protects (Sun et al., 2025); we add aligned paraphrase and audit the treatment.

3 Task and Threat Model

Cutoffs and the delivered artifact. A series comprises ordered volumes $v_1, \ldots, v_N$. At cutoff t, the reader has read $D_{\leq t} = v_1 \cdots v_t$; the cutoff follows reading order, not story time. The delivered artifact W_t includes page text, links and backlinks, titles and addresses, shared navigation and any search index, group and place pages, and source views opened by citations. All these surfaces belong to the evaluation scope. Unrequested later information is in scope; deliberately opening a later cutoff is outside the threat model.

Facts and assertions. A gold fact g is a selected character statement with reveal volume $r(g)$, the earliest volume whose text suffices to establish it. Eligible gold facts are $G_t = \{g : r(g) \leq t\}$; assessed later facts are $F_t = \{g : t < r(g) \leq N\}$. The latter are a finite probe set, not an exhaustive spoiler inventory. Story-level truth, support from eligible text and citation sufficiency are distinct properties: a true assertion supported only by later text violates the cutoff. Support must preserve attribution and modality: *A believes B is her father* does not establish *B is A's father*.

Threats. We distinguish three channels. *Construction dependence*: processing volumes beyond t shapes the early artifact through alias merges, canonical names or pages created for later characters, even when rendered claims are filtered. *Published disclosure*: a delivered surface directly exposes later content, such as a name in shared navigation. *Parametric contribution*: the generator supplies memorized information about the work. An observed disclosure alone cannot identify its channel; experiments therefore vary what each system can access (Figure 6).

4 The Probe: NARRATIVEWIKI

NARRATIVEWIKI constructs a graph of evidence-bearing claims with volume intervals and renders one wiki per cutoff (Figure 2). Claims are filtered to evidence volumes $\leq t$ before page assembly and prose generation. Filtering precedes generation because removing explicit later facts from generated text can leave implications intact.

Ingestion and entities (stages 1–2). Volumes become ordered paragraphs with stable citation identifiers such as `v01:c06:p0003`. Lexical mining identifies capitalized n-grams, honorifics and vocatives; the model types them as characters, groups, places, events or objects. Surface forms are clustered by exact match, fuzzy match and whole-word subset, followed by model adjudication of ambiguous pairs using passages containing both names. A bare name is not automatically merged with a fuller name first appearing later. Shared first names are assigned volume by volume to the locally dominant full name. Entities use the fullest title occurring in their first volume, and surface forms retain their first-occurrence volume.

Claims with evidence (stage 3). Three extraction passes share a schema: subject, predicate, object or value, qualifier, first-evidence volume (`first_vol`), verbatim quotes and paragraph identifiers. A chapter pass covers every paragraph in overlapping spans and extracts scenes, participants and summaries backed by up to four paragraphs. A mention pass reads windows around each character's mentions; a long-context pass reads all their paragraphs in a volume for attributes and background. Claims are retained only when the quote

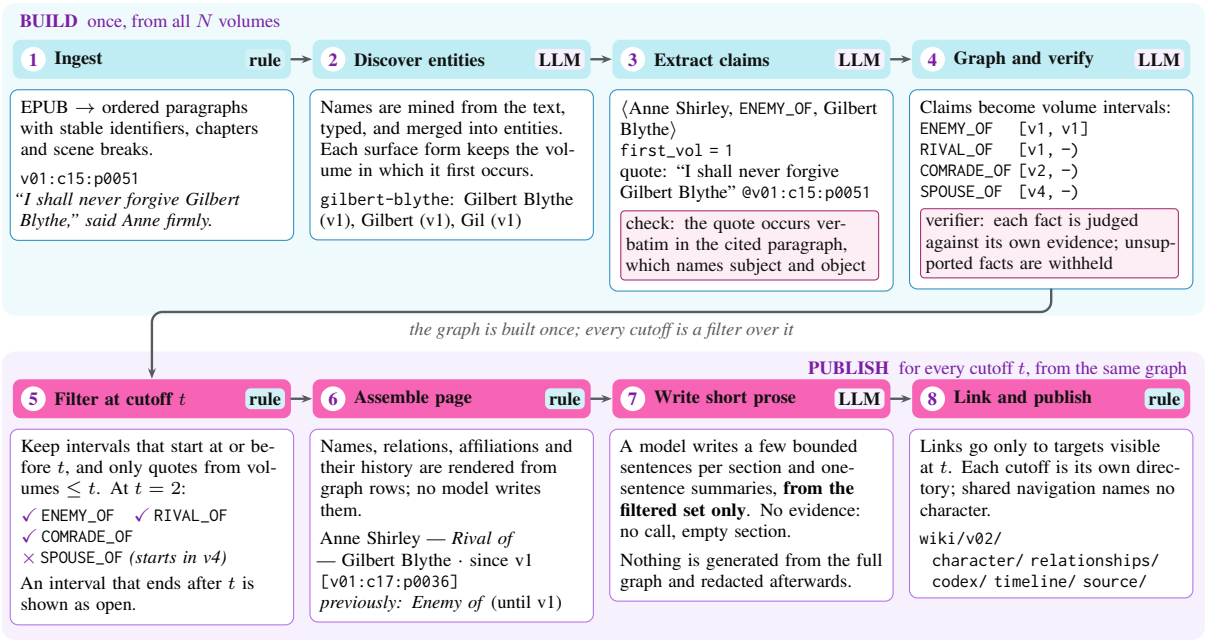


Figure 2: The NARRATIVEWIKI pipeline on a stored fact. BUILD converts all volumes into identified paragraphs, entities and claim intervals accepted by the verifier. PUBLISH assembles each cutoff's wiki from eligible intervals and quotes. Stages marked LLM call the generator; other stages are deterministic. Validation gates check verbatim evidence and fact-level support.

occurs verbatim in its cited paragraph, the subject is named in or beside that paragraph (and, for twelve two-person relations, the object in it), and relation objects resolve to known entities.

Intervals and verification (stage 4). Claims become volume intervals. Conflicting values of a single-valued attribute within one volume are arbitrated as extraction conflicts; values in different volumes are treated as story changes, closing the earlier interval when the later one opens. Near-duplicate values are clustered with BGE-M3 embeddings (Chen et al., 2024). A verifier judges each assembled fact against its cited evidence in a separate call; rejected facts are withheld. On 81 constructed minimal-edit items the verifier reaches 0.989 balanced accuracy (MiniCheck: 0.718), and 0.750 on 12 harder items of embedded speech (Appendix K).

Cutoff assembly and publication (stages 5–8). Deterministic assembly selects intervals opening by t (Algorithm 1, Appendix D). Each quote is filtered by its own volume; intervals closing after t appear current to conceal future changes; and qualifiers must be supported no later than the interval's start. Names, attributes, affiliations and relations are rendered directly from graph rows. The model writes only short prose from filtered facts, with sentence limits per section (Appendix D), and is instructed to identify the input lines supporting it. Line keys select source paragraphs; returned paragraph IDs are normalized during citation resolution. Sections without evidence remain empty without a model call. String matching adds links using surface forms revealed by t. Cutoffs occupy separate directories, shared navigation names no characters, and no site-wide search index is shipped because it would span cutoffs.

Build scope. One graph is built from all volumes. Additional cutoffs require no extraction, structured content is rendered without model calls, and prose is generated once per distinct eligible fact set. This efficiency retains construction dependence (Section 3), which we assess against prefix-only builds (Section 6.5). Every model stage, including verification, uses the generator specified in Section 5.

5 Experimental Setup

Corpus and gold facts. We evaluate the first five novels in publication order of L. M. Montgomery's Anne series, from *Anne of Green Gables* to *Rainbow Valley*, using Project Gutenberg editions: 428,122 words, 8,489 paragraphs and 184 chapters (Appendix C). The gold inventory contains 111 facts about 43 characters, each with a reveal volume and an evidence paragraph. It was

drafted from the source text without access to system outputs; eligible counts at each cutoff appear in Table 6. For 85 of the 111 facts the cited paragraph suffices alone; the other 26 (23 established in part, three not) need text beyond that paragraph and are retained unchanged (Appendix B). Recall measures coverage of this selected inventory rather than all facts in the novels.

Systems. All systems use Gemini 3.6 Flash (Vertex AI, temperature 0.4, `thinking_budget` 0). Comparison systems share a character-page template, requested sentence limits and citation format. A cast character receives a page at t if an eligible gold fact concerns them or their full name occurs in $D_{\leq t}$ (14, 21, 25, 33 and 43 pages at $t = 1..5$; Table 23). **B1** (long context) supplies the complete prefix $D_{\leq t}$ to the generator. **B2** (prefix retrieval) combines BM25 and dense retrieval from a prefix-only index. **B3** uses prefix-built LightRAG (Guo et al., 2025) as the shared generator's retriever. **B4**, the graph baseline, retrieves from the full five-volume NARRATIVEWIKI graph, filtering facts at retrieval time; it controls for the graph while changing page generation and publication scope. **B5** is NARRATIVEWIKI: it discovers its own entities and also publishes other character, relationship, codex and timeline pages. We evaluate its full artifact and separately examine the cast character pages shared with the baselines. **X1** receives only the title and character name (closed book). **X2** receives all five volumes with an instruction to reveal nothing after t. These two diagnostics are excluded from best-system comparisons. Baseline settings were fixed on a separate development novel (Appendix E).

Paired text conditions. Every system is built twice, on the original and on the treated text (Appendix N). A reviewed map of 239 entries renames characters, families and local places, and the generator paraphrases each renamed paragraph, preserving facts, order and speakers. Paragraph identifiers and the mapped gold facts carry over, so rules, code and prompts are unchanged.

Metrics. We report four separate outcomes (Figure 5; Appendix B). *Recall* is the fraction of eligible gold facts stated of the correct character anywhere in W_t. *Precision* measures full support in $D_{\leq t}$ for 40 uniformly sampled atomic assertions per system and cutoff, drawn from an inventory of all delivered surfaces (Min et al., 2023). Assertions are judged in their page context: a true scene line on a relationship page is supported there only if both characters participate in a shared episode of that scene (the *scene rule*); we also record whether each assertion is true as a statement. *Citation sufficiency* is the fraction of sampled assertions fully established by their cited paragraphs alone; uncited assertions count as insufficient. *Disclosure* searches every page for every assessed later fact using three lexical routes. Retrieved candidates are labeled *stated*, *implied*, *uncertain* or absent; stated and implied count as disclosure. There are 220 fact–cutoff assessments per system at $t = 1..4$; disclosure is undefined at $t = 5$. We measure the candidate search's recall on seeded disclosures and add two complementary checks that do not use it: later-only support among sampled assertions, and names in W_t that occur in the full entity index but not in $D_{\leq t}$.

Annotation. Labels come from LLM coding agents that read the evidence under written rules (Appendix B): OpenAI Codex and Claude Opus 5.5 produced first-pass labels, and Claude Opus 5.5 finalized every reported label, including the gold facts. The pair verdict is the judgment that moves precision most: first-pass and final verdicts agree on 90 of 99 (Cohen's $\kappa = 0.79$), and the first-pass verdicts would give NARRATIVEWIKI a precision of 0.753 instead of 0.789. Reported values incorporate consistency audits and a review-triggered recall recheck (Appendix B). Two human reviewers independently relabeled 554 original-text items, all gold facts and 10% samples of the recall, precision and citation labels, without seeing the AI answers (κ from 0.39 on precision to 0.83 on recall; Table 5).

Statistics. We report unweighted means over cutoffs with sample standard deviations; medians, ranges and paired differences appear in Tables 11 and 12. Each system and condition is built once, so standard deviations describe variation across cutoffs.

6 Results

6.1 RQ1. Without the text, memory writes the wiki; without memory, recall needs reading

Title and name alone recover two-thirds of the gold facts. Given only the title and a character's name, the generator recovers 0.683 of the eligible gold facts (0.826 at $t = 1$), states 4 of 220 assessed

System	Recall ↑ treated	Recall ↑ orig.	Recall ↑ diff.	Precision ↑ treated	Precision ↑ orig.	Citation suff. ↑ treated	Citation suff. ↑ orig.	Later facts /220 ↓ orig.	Later facts /220 ↓ treated	Distinct assertions	Tokens in / out (M)
B1 Long context	**0.845**$_{\pm 0.078}$	**0.848**	−0.003	0.920$_{\pm 0.060}$	0.929	0.530$_{\pm 0.239}$	0.658	0	0	1,821	69.23 / 0.37
B2 Prefix retrieval	0.784$_{\pm 0.052}$	0.834	−0.050	0.950$_{\pm 0.025}$	0.960	0.885$_{\pm 0.045}$	0.774	0	0	1,718	**3.42 / 0.37**
B3 LightRAG	0.470$_{\pm 0.074}$	0.536	−0.067	**0.960**$_{\pm 0.038}$	0.950	0.794$_{\pm 0.108}$	0.765	0	0	1,404	5.08 / 1.01
B4 Graph baseline	0.704$_{\pm 0.057}$	0.761	−0.057	0.940$_{\pm 0.034}$	**0.965**	**0.895**$_{\pm 0.037}$	**0.915**	0	**2**	1,701	+1.11 / 0.36
B5 NarrativeWiki	0.843$_{\pm 0.017}$	0.802	**+0.041**	0.770$_{\pm 0.065}$	0.789	0.625$_{\pm 0.087}$	0.512	**1**	**4**	**6,970**	33.13 / 1.86
X1 Closed book	0.003$_{\pm 0.007}$	0.683	**−0.680**	0.145$_{\pm 0.037}$	0.814	–	–	**4**	0	1,526	0.06 / 0.15
X2 Full text	0.744$_{\pm 0.063}$	0.814	−0.070	0.845$_{\pm 0.045}$	0.910	0.475$_{\pm 0.088}$	0.625	**8**	**23**	1,829	98.67 / 0.37

Table 1: Main results on the original text and on the treated text (names replaced, paragraphs paraphrased), with unchanged code, prompts and gold facts. Rates are unweighted means over $t = 1..5$; treated rates carry sample standard deviations (original-text distributions: Table 11). Diff.: treated minus original, before rounding. Later facts: stated or implied disclosures of the 220 assessed fact–cutoff pairs, $t = 1..4$. Distinct assertions are means per cutoff (original text); tokens cover all five cutoffs (B4: page generation added to NarrativeWiki's build). X1 and X2 are diagnostics; X1 cites nothing. Colours follow one grammar throughout the paper: cyan, best in class; violet, the headline trade-off (inventory, cost, the one recall gain under treatment); magenta, failure or disclosure.

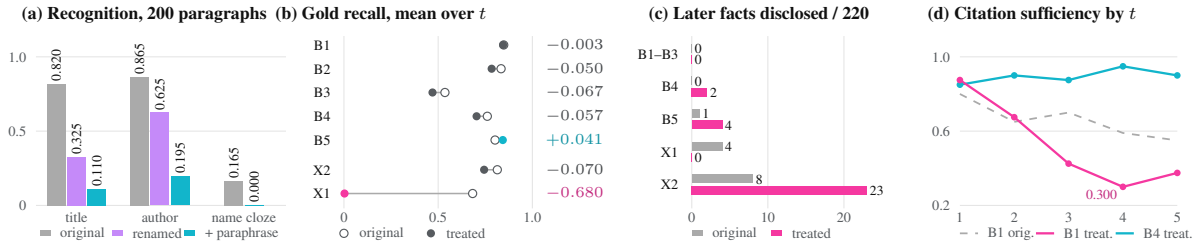


Figure 3: The renamed-and-paraphrased control (RQ1, RQ2). (a) Renaming alone leaves the work recognizable; paraphrase hides most of it. (b) Recall without text collapses; recall from text survives. (c) The instruction to withhold later volumes (X2) fails more often once the work is unrecognizable. (d) Long-context citations lose their volume on treated text; B4 on NarrativeWiki's graph does not.

later facts and reaches sampled precision 0.814 against text it never received. On a well-known work, a score is not evidence of reading.

Renaming leaves the author recognizable; paraphrase hides the book. Title identification falls from 0.820 to 0.325 after renaming alone and to 0.110 after paraphrase (Figure 3a). Renaming alone, a common control (Gupta, 2025; Zhang et al., 2024), leaves the author identifiable in 0.625 of paragraphs: the prose carries the work, not only its names.

Recall without text collapses; recall from text survives. Closed-book recall falls from 0.683 to 0.003; its single remaining match (1 of 335) is a death that also occurs in the generator's invented story. Every system given text changes far less (Figure 3b): B1 by −0.003, NarrativeWiki by +0.041, at most 0.070 (X2). The closed-book generator still writes full, invented pages (Appendix N).

Stripping recognition nearly triples instruction leaks and breaks long-context citations. X2 discloses 23 of 220 assessed later facts on treated text against 8 of 220 on the original (unique facts: 17/88 against 8/88; Figure 3c), most at $t = 1$; two treated-text assessments come from a namesake created by the name map (Appendix N). Familiar-text compliance overstated the instruction's reliability. B1's precision holds (0.920 versus 0.929), but its citation sufficiency falls from 0.875 at $t = 1$ to 0.300 at $t = 4$ (Figure 3d), as citations keep a paragraph number and lose its volume.

6.2 RQ2. Volume filters block future evidence, but modality collapse still spoils

Prefix-bounded prompting discloses no assessed later fact. B1, B2 and B3 each have no stated or implied disclosure in 220 fact–cutoff assessments on either text (Figure 6; Table 1); B4 has none on the original.

A prediction becomes a marriage. On treated text, NarrativeWiki lists the two leads as spouses at $t = 2$ and $t = 3$, a book-four fact (4 of 220: reciprocal facts at two cutoffs from one mechanism; Table 20). The cited evidence is a neigh-

Question	What the probe found	What to be careful about
RQ1 Recognition vs. reading (§6.1)	Closed-book recall is 0.683 on the original and 0.003 on treated text; systems given text lose at most 0.070.	Treatment reduces recognition without removing it (title 0.110, author 0.195); it was not proofread against its source.
RQ2 Cutoff enforcement (§6.2)	Prefix-bounded prompting discloses no assessed later fact on either text; X2 discloses 8/88 unique later facts on the original and 17/88 on treated text; NARRATIVEWIKI 1/88 and 2/88.	NARRATIVEWIKI's treated-text disclosures are one mechanism at two cutoffs; zero counts depend on a candidate search that misses many pronoun references (Table 14).
RQ3 Accumulation (§6.3)	Central-three recall 0.929 for NARRATIVEWIKI against 0.794 for long context (treated); one fact lost across cutoffs per text; 6,970 distinct assertions per cutoff.	The central/other split was chosen after the results; the comparison pages are sentence-budgeted, not length-matched.
RQ4 Context (§6.4)	0.934 of NARRATIVEWIKI's sampled assertions are true as statements, but precision in page context is 0.789; B4 on the same graph reaches 0.965 and citation sufficiency 0.915.	Page scope differs between NARRATIVEWIKI and B1–B4; precision rests on 40 sampled assertions per cell, with the lowest human agreement of all tasks (Table 5).
RQ5 Construction scope (§6.5)	0.390 of character pages at $t = 1$ differ from a prefix-built index (0.220 regrouped or absent); prefix-index wikis match pooled recall (180 of 224).	Single builds per index, so model variance is inside the differences; prefix-index wikis were labeled in one pass, on original text only.

Table 2: The paper at a glance. Each research question with its answer and the boundary that limits it. Cyan: the gain the design delivers; magenta: the failure a volume filter cannot detect.

bour's prediction in book two; extraction stored a presumed marriage, and deterministic rendering printed presumed spouses as spouses. B4, generating from the same graph, qualifies the pair as an "anticipated future marriage" without the speaker (2 of 220, implied). Once stored, the claim keeps a coarse *presumed* flag but not who spoke or that it was a prediction: a volume filter checks *when* evidence appears, not *what* it licenses (Proposition 3, Appendix I). On the original text NARRATIVEWIKI has one implied disclosure: at $t = 3$ a summary says a friend *later serves as a bridesmaid for Anne*, where the eligible passage makes Anne the friend's bridesmaid.

Zero counts are bounded, not certified. Counting uncertain verdicts raises the original-text count of B1–B5 from 1 of 1,100 to 9 of 1,100 assessments (NARRATIVEWIKI: five). The candidate search retrieves 0.845 of seeded paraphrases that name the character, and fewer when only a pronoun refers to it (Table 14); no sampled assertion at $t \leq 4$ in NARRATIVEWIKI or B1–B3 is established only by later volumes (NARRATIVEWIKI: 0 of 159).

6.3 RQ3. Accumulated claims grow with the protagonists; budgeted rewrites forget

Central-character recall rises as budgeted pages decay. Comparison systems rewrite every page at every cutoff in one call under a shared sentence budget (Appendix E); NARRATIVEWIKI renders every eligible claim, so Anne Shirley's page grows from 1,813 words to 3,439, while B1's never exceeds 1,157 (Figure 4b). On the three characters with the most gold assessments, NARRATIVEWIKI has the highest recall of any system on treated text (0.929, against B1's 0.794) and, by a smaller margin, on the original (0.894 against 0.837). On treated text its recall on them rises from 12/14 at $t = 1$ to 36/38 at $t = 5$, while B1's falls from 14/14 to 29/38 (Figure 4a), as a fixed budget predicts (Proposition 2).

Stored claims are not forgotten. A fact conveyed at t and lost at $t + 1$ occurs once for NARRATIVEWIKI on each text; every other system loses 4–19 on the original and 11–24 on treated text (Figure 4c). B4 reads the same graph and loses 12 and 14: the stability comes from rendering stored claims (Proposition 1), not from the graph alone.

One build publishes 3.8 times the distinct assertions. On treated text NARRATIVEWIKI's recall is close to long context's (0.843 versus 0.845), and it is the only system whose recall did not fall under treatment ($+0.041$). On the original text, one build publishes a mean of 6,970 textually distinct assertions per cutoff, against at most 1,821 for B1–B4 (Table 25).

6.4 RQ4. Statements stay true in isolation, but lose support on pair pages

Coverage trails on minor characters. The central-character gain does not carry to the average: on the original text NARRATIVEWIKI trails long context (0.802 versus 0.848) and prefix retrieval (paired difference -0.032; Table 12), which uses

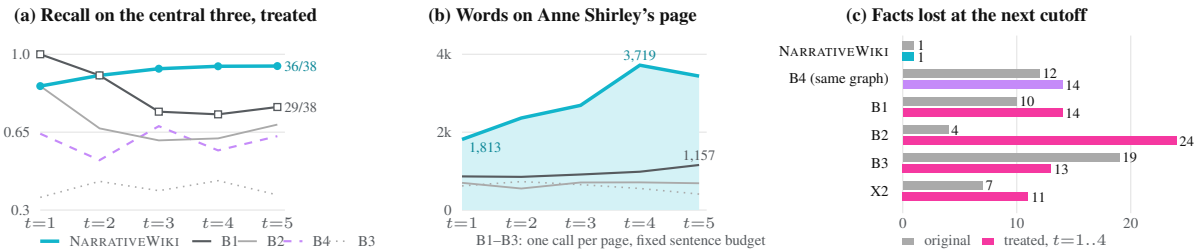


Figure 4: Where accumulation helps (RQ3). (a) On the three most-documented characters NARRATIVEWIKI's recall rises with the prefix while budgeted rewrites fall. (b) NARRATIVEWIKI's page for Anne grows with the story. (c) Facts conveyed at t and lost at $t+1$: B4 reads the same graph but rewrites each page, so stability comes from rendering stored claims.

a tenth of the input tokens. The deficit lies with minor characters (Table 24).

Placement, not invention, lowers precision. Whole-artifact precision is 0.789 for NARRATIVEWIKI (0.628 under the stricter line rule), yet 0.934 of its sampled assertions are true as statements. On cast character pages, the matched population, NARRATIVEWIKI reaches 0.886 (31/35), versus B1's pooled 0.930 (185/199). The gap concentrates on relationship pages (46% of its assertions): 0.971 (99/102) are true as statements but 0.706 (72/102) are supported in page context, because scene lines naming both characters reach the pair's page even when the two share no episode (Figure 8a, Appendix G; Proposition 4).

The same graph supports precise, well-cited pages. Citation sufficiency is 0.512 for NARRATIVEWIKI against 0.658 (B1) and 0.774 (B2), and 0.534 (102/191) against 0.668 (131/196) and 0.802 (154/192) on sampled assertions that carry a citation, so omission does not explain the gap. B4, which writes cast pages from facts retrieved from NARRATIVEWIKI's graph, has lower recall (0.761) but the highest original-text precision and citation sufficiency (0.965, 0.915): much of the gap arises where claims are selected, published and cited.

6.5 RQ5. Whole-series entities carry later cast structure into early pages

For the reported build, 0.390 of character pages at $t = 1$ and 0.089 at $t = 4$ have no prefix-built character with exactly their set of names observed by t; excluding alias-only differences, 0.220 and 0.052 are regrouped or have no prefix counterpart (Figure 8b; Table 17; first-build prefix graphs: Table 16). Wikis built on those prefix indexes match the full build's pooled recall (180 of 224 each, $t = 1..4$) and state no later fact, though both keep the implied bridesmaid disclosure at $t = 3$ (Table 18): prefix-built indexes remove the dependence at no cost in pooled recall. Early-volume prompts also carry later aliases (Appendix Q, D3), a channel these comparisons do not isolate.

7 Discussion

One cause behind both sides. A claim replaces its passage as the generator's input (Table 2). Because claims accumulate, a page can grow and keep what it said, which a page rewritten under a fixed budget cannot (Propositions 1–2, Appendix I). Because a claim keeps a short quote rather than its discourse, the filter tests the one property it retains exactly, the volume of its evidence; speaker, mood, namesake and shared episode are held weakly or not at all (Propositions 3–4). Both effects sit at publication, not only at extraction.

Where the trade-off should pay. Accumulation matters where rewriting each page is costly: series longer than one context window, and characters whose record outgrows a page budget. The failures are repairable without giving up accumulation: carry speaker and modality in the claim schema, validate entity groupings against the prefix, and check each assertion in its page context. Appendix J turns the audits into six reporting practices.

8 Conclusion

Decomposing a narrative into dated claims filtered at the reader's cutoff trades context for accumulation: pages grow and keep what they said, while failures arise on evidence the filter permits. Our gold facts, labels and renaming pipeline make both sides testable on longer works.

Limitations

One series, one generator, one build per cell. The evaluation covers five volumes of one English public-domain series and one generator, so run-to-run variance is unknown, and cutoffs share characters and evidence rather than being independent samples. Precision and citation sufficiency rest on 40 sampled assertions per cell. All five volumes fit in the generator's context window, which favors direct prompting; longer series, other genres, languages and generators would test where accumulation pays (Section 7).

The Anne series informed development and is not a held-out test set. The reported NARRATIVEWIKI is the second of two builds, revised after the first was labeled; Table 21 reports both under one rule, and the revision moved metrics in both directions. The baselines, configured on another novel, were not retuned. The prefix-index wikis of Section 6.5 were built on the original text only and labeled in one pass.

An unblinded LLM finalized the labels; human checks cover original-text subsets. The finalizing annotator, Claude Opus 5.5, knew which system produced each output and assisted in building the system. Two human reviewers relabeled original-text gold, recall, precision and citation items (Table 5); disclosure and treated-text labels have no human check, and agreement is lowest for precision and citation sufficiency. Some gold evidence paragraphs establish their facts only in part (Section 5).

Lexical candidate retrieval under-samples pronoun-mediated disclosures. Disclosure is assessed for a selected inventory of later facts through candidate retrieval, validated on seeded fixtures written by the same LLM annotator (Table 14). The line between an implied disclosure and an eligible prediction rests on the annotator. A zero count therefore does not certify spoiler-free output, and NARRATIVEWIKI's larger output leaves more room for missed disclosures.

Accumulation gains are measured against sentence-budgeted, not length-matched, pages. The comparison systems write each page in one call under a sentence budget, while NARRATIVEWIKI renders every eligible claim; an append-only or length-matched baseline would separate graph representation from page length, and was not run. The central/other split was chosen after the results were known, and facts lost across cutoffs also count labeling inconsistencies between adjacent cutoffs. Identity-dependence rates compare single builds. B4 differs from NARRATIVEWIKI in page scope and generation as well as citation attachment, and neither the account of Section 7 nor the proposed repairs have been tested by ablation.

Renaming and paraphrase reduce recognition without removing it. The treated text was not proofread against its source: some paragraphs keep inflected source names, the name map creates namesakes, and one paraphrase changes a fact (Appendix N). Its fidelity to the gold facts was checked by an unblinded LLM. A work published after the generator's training cutoff would offer a complementary test (Karpinska et al., 2024).

The representation is coarse and readers were not studied. Cutoffs are whole volumes; identity links are not dated; attribute intervals track explicit replacements rather than implicit aging; a value that changes within one volume is arbitrated as an extraction conflict (Appendix Q); and cross-claim inference is limited (Appendix O). We do not measure readers' ability to find information or their experience of spoilers. External search engines and deliberately opened later cutoffs are outside the audited artifact.

Ethical Considerations

The evaluation uses public-domain Project Gutenberg editions. The accompanying materials include code, identifiers, gold facts, AI labels and human-review summaries, and exclude copyrighted development texts; no reported measurement comes from those development series. No reader-study participants were recruited. Claude Opus 5.5 finalized the AI labels, with human review of selected original-text items as described in Section 5 and Appendix B.

AI coding assistants contributed to implementation, evaluation tools, drafting and the vector figures, which are drawn from stored data and measured values. The authors reviewed these materials and remain responsible for their content. Cutoff-bounded references are intended to assist readers, although generated summaries can substitute for reading and can inadvertently disclose later events. Released pages concern public-domain works. The paper itself contains plot spoilers for the Anne se-

ries.

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Appendix Contents

The appendices provide annotation rules, implementation details, complete results, a formal account of the trade-off, reporting recommendations and the prompt and defect record.

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A Comparison with Prior Work

Table 3 compares NARRATIVEWIKI with closely related systems and evaluations. The entries follow each cited paper's description of its task, output, and evaluation. Among these works, none combines a published reference artifact, a reader-selected volume cutoff, and evaluation of atomic assertions across the delivered pages.

B Annotation Rules and Metric Definitions

Annotators and workflow. Labels were produced by LLM coding agents working interactively in the project repository under the written rules of this appendix, not by a fixed prompt: for each item the agent read the fact or assertion with the eligible source paragraphs and recorded a label in a released label file. First-pass labels came from two agents, OpenAI Codex and Claude Opus 5.5; Claude Opus 5.5 finalized every reported label. The model version behind the Codex agent was not recorded. The prefix-index builds of Table 18 received a single pass.

How each metric is computed. Let A_t be the atomic assertions of W_t, $S_t \subset A_t$ the sampled atoms (40 drawn uniformly, with enumeration frame errors removed), $c(a)$ the set of paragraphs cited for atom a, $c_t(a) = c(a) \cap D_{\le t}$ its eligible part, and $\mathbf{1}[\cdot]$ the indicator. For an assessed later fact $g \in F_t$, $d_t(g)$ is the verdict of reading every candidate sentence of W_t for it.

$$
\begin{aligned}
\mathrm{M4}_t &= |P_t| \,/\, |F_t| \qquad (t \le 4),\\
P_t &= \{g \in F_t : d_t(g) \in \{\text{stated}, \text{implied}\}\},\\
\mathrm{M8}_t &= \frac{1}{|S_t|} \sum_{a \in S_t} \mathbf{1}\big[\sigma_t(a)\big],\\
\sigma_t(a) &\equiv c_t(a) \neq \varnothing \wedge c_t(a) \models a,\\
\mathrm{CITE}_t &= \frac{|\{a \in A_t : c(a) \neq \varnothing\}|}{|A_t|},\\
\mathrm{M5} &= \frac{|L|}{\sum_{t \le 4} |S_t|},\\
L &= \{(t,a) : 1 \le t \le 4,\ a \in S_t,\\
&\qquad D_{\le t} \not\models a,\ D_{\le 5} \models a\},
\end{aligned}
$$

where $c_t(a) \models a$ means the eligible cited paragraphs, read alone and jointly, establish every element of a (*sufficient* below); cited paragraphs after the cutoff are ignored, so an atom citing only later paragraphs is insufficient. The disclosure count of Table 1 is $\sum_{t=1}^{4} |P_t|$; the inclusive count adds the facts whose verdict is *uncertain*. A system's value is the unweighted mean over the cutoffs measured, $\bar{M} = \frac{1}{|T|} \sum_{t \in T} M_t$. A paired difference between systems A and B is $\Delta = \frac{1}{|T|} \sum_{t \in T} \big(M_t^A(E_t) - M_t^B(E_t)\big)$, with E_t the page entities both have at t; its standard deviation, median, lowest and highest value are those of the per-cutoff differences. The recall gap relative to the closed-book diagnostic is $\bar{\mathrm{M}}1 - \bar{\mathrm{M}}1^{X1}$; it does not isolate the contribution of reading from memory. The seeded search recall is the share of seeded disclosure sentences returned as candidates. M5 assesses a uniform assertion sample, whereas M4 uses fact-directed searches across the delivered artifact; their denominators and detection coverage differ.

Unit. An atomic assertion states one predicate about one subject, and an object for relations. Its interpretation includes polarity, modality, and attribution to the speaker or believer when the narrative does not establish it as true.

Support labels (M3). *Supported*: the text through the cutoff establishes the assertion with the stated attribution and modality. *Partial*: part of it is established, or a transient state is presented

Work	Task	Output artifact	Cutoff	Construction scope	Evaluation unit
Jones et al. (2018)	spoiler-free wiki browsing	archived fan-wiki revision	episode air date	human-written revisions saved before the cutoff	revision selected for a requested date
CHIRON (Gurung and Lapata, 2024)	character representation	character sheet	story snippet	story so far	generated statement; masked-character prediction
TimeChara (Ahn et al., 2024)	point-in-time role-play	dialogue answers	story moment	not applicable	response to one interview question
Living the Novel (Huang et al., 2025)	conversational agent	dialogue	narrative present	whole novel, retrieval gated at t	character dialogue; user study
STORM (Shao et al., 2024)	article writing	article	none	web search, target article excluded	outline headings; article sentence citations
NarrativeFact-Score (Jeong et al., 2025)	summary evaluation	factuality score and feedback	none	whole narrative	atomic fact of a summary
STORYSUMM (Subbiah et al., 2024)	faithfulness benchmark	labels	none	not applicable	summary sentence
NS-ST-GraphRAG (Lin, 2026)	question answering	answers	temporal scope of a query	whole work, subgraph selected at query time	answer to a question
SearchWiki (Singh et al., 2026)	information seeking	linked wiki for an agent	none	whole corpus	downstream answer
NARRATIVEWIKI (this work)	reference generation	linked wiki per cutoff	reading volume	all volumes, rendered from facts filtered at t; compared with prefix-only builds	atomic assertion of a delivered page; later gold fact against every page

Table 3: Comparison with prior work by task, artifact, cutoff, construction scope, and evaluation unit.

as a lasting trait. *Unsupported*: nothing through the cutoff establishes it. *Uncertain*: the annotator cannot decide from the eligible text. The annotator may read surrounding eligible context but never text after the cutoff. Only *supported* counts. Each atom is judged in the context of the page on which it appears; its factuality in isolation is also recorded (*true as a statement*). An error introduced by the enumeration tool, such as assigning a quotation to the wrong speaker, is a frame error and is excluded from the denominator (at most 1 per cell).

The pair rule. Relationship pages display lines from scene summaries as shared scenes. A factually true line is supported *on that page* only if both characters are present and participate in at least one episode of the scene summarized. Being mentioned, quoted, written to, remembered, dead, or watching from a distance does not count. This *scene rule* is the standing one. Under the stricter *line rule*, reported as a sensitivity value, the pair must take part in the episode that the sampled line itself describes. An earlier rule that accepted a pair present anywhere in the chapter was retired because it conflated shared scenes with shared chapters. The first-pass labels of the first and the reported build had used different pair rules, so Claude Opus 5.5 re-read every sampled relationship line from the first build, reported build, and two prefix builds against the chapter text. The annotator recorded verdicts under both the scene rule and the line rule. These contextual verdicts are combined with each build's factuality labels to compute the page-context precision reported here.

Reveal volume. The earliest t whose prefix $D_{\leq t}$ establishes the fact without later text; foreshadowing that a first-time reader could not resolve is not sufficient. Each later fact's reveal volume was checked against the earliest statement in the text by a keyword scan of every earlier volume followed by a read; the check moved 2 of 90 drafted facts to an earlier volume.

Conveyed (M1). A gold fact is conveyed at cutoff t when any page of that cutoff's wiki (a character, relationship, codex or timeline page) states its core proposition of the right character (the same subject, the same event, relation or attribute, and the same object) in any wording. A conjunction of two propositions needs both (*a newspaper man and a grandson of the schoolmaster*). An incidental qualifier may be missing (*four years ago*, *at*

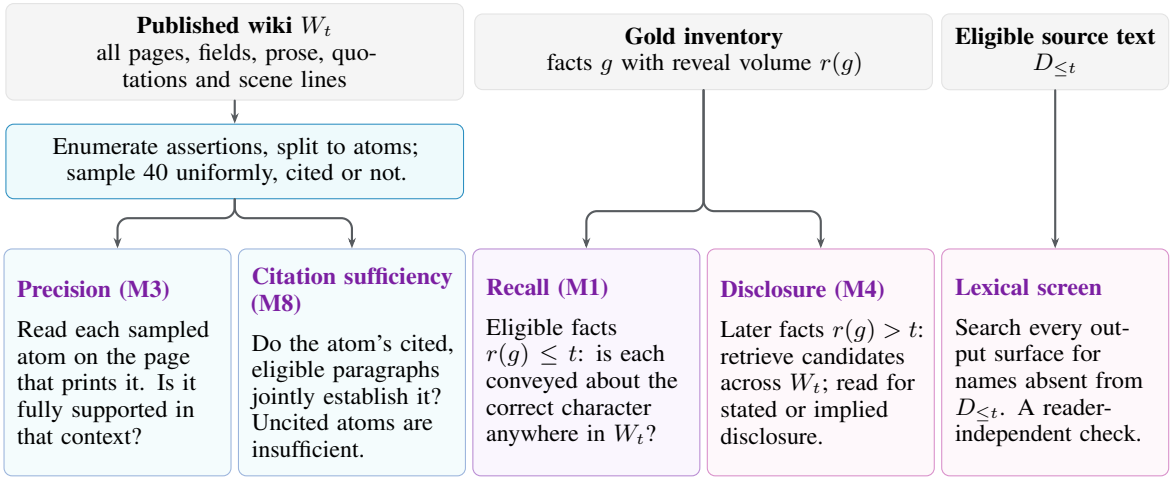


Figure 5: The evaluation protocol, applied identically to every system's wiki at every cutoff. Cyan: support for the wiki's assertions. Lavender: coverage of information available to the reader. Pink: checks for later information reaching the reader.

ID	Definition
M1	Recall: eligible gold facts conveyed about the correct character anywhere in W_t, divided by $\|G_t\|$.
M3	Precision: fully supported sampled atoms divided by sampled atoms across all surfaces, cited or uncited; enumeration frame errors are excluded.
M4	Disclosure: assessed later facts stated or implied anywhere in W_t, divided by $\|F_t\|$; undefined when $F_t = \varnothing$.
M5	Later-volume content: sampled atoms not established through the cutoff but established by the full five-volume text, divided by sampled atoms at $t = 1, \ldots, 4$.
M8	Citation sufficiency: sampled atoms established by their cited, eligible paragraphs, divided by all sampled atoms; uncited atoms are insufficient.
M12	Verifier errors: unsupported candidates admitted and supported candidates rejected on a challenge set.
M13	Cost: model calls, input tokens, and output tokens of the reported build.

Table 4: Metrics reported in this paper. IDs retain the historical measurement-ledger numbering to match archived results and labels; measures outside the reported set leave gaps.

Christmas, a color or a material); a wrong or missing core element may not: who did it, to whom, or the number when the number is the fact (*Una is ten*). The same proposition asserted of a different character does not count (where the statement stands does not matter, whom it is about does), and an engagement does not convey a marriage. We formalized this rule across all systems after the recall audit in Section 5 found inconsistent application in the first pass. Labels had been carried forward across cutoffs when similar wording persisted; a fact marked absent early was therefore not reassessed when a later wiki paraphrased it. The audit rechecked every gold fact in every system using one search pattern per fact and changed 108 (96 to conveyed, 12 to not conveyed) labels; the reported build of NARRATIVEWIKI and the graph baseline on it were labeled afterward with the same patterns.

Disclosed (M4). *Stated*: a page asserts the later fact. *Implied*: a page asserts something from which a reader would infer it. *Uncertain*: a page asserts something adjacent that is available at the cutoff and that a reader could, but need not, read toward the later fact. *Absent*: no candidate sentence was found, or every candidate read was about something else. Stated and implied count as disclosure. Candidates are found by three lexical routes: vocabulary that first occurs after t, overlap with the content words of the fact, and marriage, death and birth words near the character's name.

Citation sufficiency (M8). For a sampled atom that the page cites, the annotator opens every paragraph in the citation list of that line or section (only paragraphs at or before the cutoff count) and asks whether they establish the atom without any other text. *Sufficient*: every element of the atom is in the cited paragraphs, jointly if need be; a relation is sufficient when the paragraph states it of a person the reader can identify, even if the page's canonical surname is given elsewhere. *Partial*: the core is there and a stated element is not (an added qualifier, a trait inferred from behavior). *Insufficient*: the supporting paragraph is not in the list, the list points

to another volume or person, or the atom is itself unsupported. An assertion with no paragraph citation is insufficient; a link to a whole chapter is not a paragraph citation. Only *sufficient* counts. One correction was made after labeling. The tool that enumerates assertions did not attach to a timeline line of the reported build the *Sources* line printed under it, so the 22 sampled timeline atoms had been labeled as uncited. Their citations were read from the delivered pages and judged by the rule above: 11 are sufficient. Counting them as uncited would give NARRATIVEWIKI a citation sufficiency of 0.457 instead of 0.512, and a share of assertions with a paragraph citation of 0.840 instead of 0.930.

The precision audit. Citation assessment for M8 revealed comparison-system atoms previously labeled unsupported or uncertain that were established by the text, sometimes by the page's own citations. The first pass searched for a character's name near the fact, missing pronoun references, and examined only the first six cited passages. All 119 comparison-system atoms not labeled supported were re-read against the best-matching paragraphs of the complete eligible text, and 34 were corrected to supported. The atom-level negatives of NARRATIVEWIKI's first build were re-read the same way and none changed. Labels already marked supported were not re-audited for any system.

Two-reviewer agreement. Two human reviewers each labeled the complete review packet (554 original-text items: every gold fact and the samples of recall, precision and citation labels listed below), giving two separate sets of final labels. Table 5 recomputes their agreement on the original-text validation tasks. Cohen's κ is reported separately for gold evidence, recall, precision and citation sufficiency, since these tasks have different questions and label spaces. Across the 554 reviewed items, all assessed decisions agree on 458; at least one decision differs on 96. These values describe concordance between the reviewers' final judgments, rather than agreement with the AI annotations or correctness of the annotations. Each reviewer labeled every item from scratch, without seeing the AI answers, so these judgments are not anchored on them.

Table 5 compares each reviewer's label set with the accepted AI-assisted review snapshot finalized by Claude. Its mapping to the accepted source sheet is verified; it is not independently verified as unchanged blind first-pass AI labels. Each comparison uses all 554 reviewed items. Pair placement is evaluated separately on 14 relationship-page items within the precision sample. All applicable decisions agree on 473 items for reviewer 1, with at least one disagreement on 81; for reviewer 2, the corresponding counts are 457 and 97. These comparisons measure reviewer–AI concordance, not annotation accuracy.

Reviewer provenance. Both final label sets were produced entirely by human reviewers; no model generated or proposed these decisions. Both CSVs cover the complete review packet, and their labels match the accompanying response records. A released aligned audit maps each decision to its original item and marks remaining disagreements as unadjudicated.

Joint agreement across the three label sets. Table 5 recomputes nominal Krippendorff's α for the same accepted AI-assisted reference and the two reviewers' final-label sets. All 554 items have complete labels from all three. For each task, $\alpha = 1 - D_o/D_e$ uses nominal distance (zero for identical labels, one otherwise), observed within-item coincidences, and expected disagreement from pooled category counts with the $n(n-1)$ finite-sample correction, where n is the number of pairable label values (Krippendorff, 2011, pp. 1–5). It is calculated jointly, rather than by averaging pairwise kappas. Pair placement is again assessed separately within the precision sample. The coefficient measures concordance among the AI labels and two human label sets; it is not an all-human reliability estimate or an accuracy measure. The recomputed coefficients match the summary distributed with the label files. This calculation reproduces decisions; it does not adjudicate disagreements or change system scores.

Earlier human review. Claude Opus 5.5 finalized all reported AI labels. Before the two-reviewer check, one human reviewer examined and accepted an AI-checked sheet with answers and reasons visible; this pass yields no agreement estimate. It assessed the labels against the source text and the rules above. Sets of at most 500 items are checked in full; larger sets use a uniform random 10% sample. Each set is checked separately:

- gold facts, 111 items (at most 500): every fact and its evidence paragraph;
- recall labels, 2,345 items (over 500): a sample of

		Reviewer 1 vs. 2		Reviewer 1 vs. AI		Reviewer 2 vs. AI		All three label sets		
Task	**Items**	**Agr.**	κ	**Agr.**	κ	**Agr.**	κ	α	**Mean pairwise**	**Unanimous**
Gold evidence	111	82.88%	0.55	85.59%	0.58	80.18%	0.51	0.54	82.88%	83
Recall (M1)	235	**92.77%**	**0.83**	**94.47%**	**0.87**	**89.36%**	**0.75**	**0.82**	92.20%	209
Precision (M3)	148	70.95%	0.39	77.70%	0.48	80.41%	0.50	0.45	76.35%	97
Citation (M8)	60	71.67%	0.57	71.67%	0.57	68.33%	0.49	0.54	70.56%	35
Pair placement	14	92.86%	0.86	71.43%	0.44	64.29%	0.29	0.54	76.19%	9

Table 5: Human–AI concordance on the 554 doubly reviewed original-text items. Agr.: percentage agreement; κ: Cohen's kappa per task; α: nominal Krippendorff's alpha over the accepted AI-assisted reference and both human label sets; mean pairwise agreement averages the three label-set pairs; unanimous: identical labels in all three sets. Pair placement is a subset of the precision sample, not additional items. No mixed-task coefficient is pooled. Cyan: highest agreement; lavender: lowest, the context-sensitive precision task.

235;
- precision atoms, 1,480 items (over 500): a sample of 148;
- citation-sufficiency labels, 600 items (over 500): a sample of 60;
- disclosure verdicts, 1,540 items, and the 12,689 candidate sentences behind them: not checked by a human; these are LLM labels only.

In all, 554 labels were reviewed. The sampling frame was fixed before the last labels were added: its precision atoms are those of every original-text cell then labeled, including prefix-build cells, and its citation labels are those of B1, B2 and NARRATIVEWIKI; the citation labels of B3, B4 and X2 were added later and are outside it. All of them are labels of the original text: the labels of the treated text (Appendix N) were finalized afterward by Claude and are not part of the check. The accepted answers agree with the paper's original labels on 204 of 235 recall items, 120 of 148 precision atoms and 46 of 60 citation labels. Of the gold facts, 85, 23 and 3 are established by their evidence paragraph in full, in part and not at all; 12 of those established in part are dated by a paragraph that announces the event, which is completed within the same volume, so no reveal volume moves. For the three not established, the fact holds in the volume it is dated to, but the paragraph recorded as its evidence does not carry it alone (a greeting for a household arrangement, a plan for a residence, an introduction that does not yet tie a surname to the man). In recall and precision the accepted answers are stricter; citation-sufficiency differences occur in both directions.

The recall differences had one cause: in some cells a page sentence weaker than the gold fact had been credited ("fascinated by insects" for a boy who collects toads, bugs and frogs; a fight for a hatred). The written rule already asked for the same event, relation or attribute. We therefore re-read six gold facts in every cell of every system, on both texts, with one test per fact, and changed 64 (48 to not conveyed, 16 to conveyed) recall labels. The re-read moved no system's mean recall by more than 0.021 and is in every value reported here; after it the accepted answers match 215 of 235 recall items. It covers the six facts the check surfaced, not all 111. Precision and citation labels were not changed on the sampled items alone, since that would correct a tenth of the labels under a reading the other nine tenths did not get.

Zero counts. A zero disclosure count is printed as $0/n$. Under an idealized model of k independent, representative character-level Bernoulli trials with zero events, where k is the number of characters with an assessed later fact at t (40 at $t = 1$, 12 at $t = 4$), the exact one-sided 95% upper bound is $1 - 0.05^{1/k}$: 0.072 at $t = 1$ and 0.221 at $t = 4$. Grouping by character avoids treating correlated assertions as independent. However, characters also interact, the cast is purposively selected, and candidate retrieval is incomplete. These values illustrate the effect of the sampling unit; they are not calibrated population bounds on this system's spoiler risk.

C Corpus and Evaluation Dataset

Editions. The Anne series is read from Project Gutenberg eBooks 45 (*Anne of Green Gables*), 47 (*Anne of Avonlea*), 51 (*Anne of the Island*), 544 (*Anne's House of Dreams*) and 5343 (*Rainbow Valley*), in that order (Table 6). The corpus follows the publication order of the first five Anne books (1908–1919), excluding two later-written books set

t	Book	Par.	Words	Ch.	$\lvert G_t \rvert$	$\lvert F_t \rvert$
1	*Anne of Green Gables*	1,788	101,965	38	23	88
2	*Anne of Avonlea*	1,705	90,685	30	46	65
3	*Anne of the Island*	1,759	76,061	41	66	45
4	*Anne's House of Dreams*	1,606	78,931	40	89	22
5	*Rainbow Valley*	1,631	80,480	35	111	0
	Total	**8,489**	**428,122**	**184**		

Table 6: The evaluated corpus. Paragraphs, words and chapters are per volume; $|G_t|$ is the number of gold facts eligible at cutoff t and $|F_t|$ the number of later facts assessed for disclosure at t.

Primary-matrix unit	Original	Treated
System–cutoff artifacts	35	35
Selected gold records	111	111
Cast characters	43	43
Recall judgments	2,345	2,345
Disclosure judgments	1,540	1,540
Sampled assertion records	1,400	1,400
Excluded frame errors	4	2
Evaluated precision atoms	1,396	1,398

Table 7: Size of the primary evaluation matrix, excluding earlier and prefix-build diagnostics. Counts describe records and judgments, not independent trials. Both conditions reuse the same gold inventory. Citation sufficiency uses the sampled assertion records; X1 supplies no citations.

between them. Throughout the paper, a volume is one book; word counts use whitespace-delimited tokens in the parsed text.

Development history. The pipeline was developed on other series, which contribute no reported results. The Anne books were processed while the pipeline was still being revised, so this corpus is not a held-out evaluation. Baseline settings were chosen on George Eliot's *Middlemarch* (Gutenberg eBook 145, Books I–V) and not on the Anne series (Appendix E).

Cast. The evaluated cast contains the 43 characters with at least one gold fact. A comparison system generates a character page at cutoff t if a gold fact about that character is evidenced by t or their full name occurs in $D_{\leq t}$. Characters without a page remain in the evaluation denominators. This rule replaces an earlier rule that matched any single word of a name and generated baseline pages for characters not yet introduced (Appendix L).

Dataset size and units. The source corpus contains 428,122 whitespace-delimited words in 8,489 paragraphs and 184 chapters. The evaluated inventory contains 111 selected facts about 43 characters. Table 7 distinguishes these unique records from repeated judgments. Each original or treated condition contains the same seven systems at five cutoffs; renaming and paraphrasing reuse the inventory and reveal dates rather than adding an independent corpus. Recall labels judge eligible fact–cutoff pairs; disclosure labels judge later fact–cutoff pairs. Assertions are a separate system-generated population sampled for precision and citation sufficiency. Repeated cutoffs share facts and characters.

What a record contains. The gold YAML groups facts by character. Each fact records a claim and a paragraph identifier; the identifier encodes its evidence volume, chapter and paragraph. Optional keyword lists assist disclosure search, while reader and note fields record provenance. The stored YAML retains historical later-volume entries, but this evaluation selects evidence volumes one to five only. Below is a minimal excerpt from an existing record, with provenance metadata omitted:

```
series: anne
characters:
  Matthew Cuthbert:
    facts:
      - claim: "Matthew is the shyest man alive."
        evidence: v01:c01:p0004
```

Easier example: a direct statement. The cited paragraph describes Matthew as "the shyest man alive" and says that he "hated to have to go among strangers". The human-accepted gold check (Appendix B) labels this record fully supported. The character and description occur together, so matching the proposition requires neither a cross-paragraph identity link nor a conjunction. Its evidence volume is one: it is eligible at every evaluated cutoff. This is the narrator's characterization of Matthew, rather than a claim about his current life status.

Harder example: conjunction and local coreference. Owen Ford's record reads: "Owen Ford is a newspaper man and a grandson of the schoolmaster who built the house", with evidence `v04:c22:p0019`. Miss Cornelia's report includes "His name is Owen Ford", "a newspaper man" and "a grandson of the schoolmaster who built this house". The human-accepted gold check labels the record fully supported. Interpreting the pronouns requires linking them to Owen; recall requires both the occupation and the family/house relation. A generated state-

ment that supplies only his occupation does not convey the full record. At cutoff three the fact is a disclosure probe; at cutoff four it becomes eligible. The example illustrates the existing conjunction rule, not a measured difficulty grade or a newly assigned annotation.

An evidence boundary case. The gold claim "Anne marries Gilbert Blythe" points to `v04:c04:p0000`, which begins "Anne wakened on the morning of her wedding day". That paragraph alone neither names Gilbert nor establishes the completed marriage. The human-accepted gold check therefore marks it partial, even though the fact is assigned to the correct volume. This example distinguishes volume-level support from paragraph-level evidence sufficiency; the recorded evidence defect remains part of the inventory, as reported in Appendix B. These illustrative cases do not constitute new labels, a measured easy/hard split, or a repair of the gold evidence.

Review population and release scope. The two-reviewer check covers 554 original-text items: 111 gold-evidence, 235 recall, 148 precision and 60 citation checks, sampled from the labels that existed when the sample was drawn (Appendix B). Original and treated gold, per-system judgments, sampled assertions, candidate witnesses, the two reviewers' final label files and recomputed agreement form the evaluation materials. The reviewer files, original response records and input hashes are released; the human review is described in Appendix B.

D Implementation Details

Claim schema. A claim is a record with a subject (an entity identifier, never a raw surface form), a predicate, a kind (attribute, relation or trait), an object entity or a scalar value, an optional short qualifier, a polarity (asserted, denied or presumed), the volume and chapter of its first evidence, and one or more evidence items, each a paragraph identifier with a verbatim quote of at most 240 characters. *Presumed* keeps a belief apart from a fact: a character thought dead in one volume is recorded as presumed dead there, not as dead. Polarity and qualifier are part of a claim's identity, so two observations that disagree on either never merge. A claim whose subject or object cannot be resolved to a known entity is dropped.

Ontology. Entities are typed as character, group, place, event or object; only characters get their own pages, and the other types share codex pages. There are nine attributes (age, gender, status, rank, title, nickname, origin, occupation, appearance), five trait keys (personality, motivation, skill, fear, background) and twenty relation predicates covering kinship (parent, child, sibling, relative, spouse), affect (romantic, friend, rival, enemy), roles (mentor, student, commands, serves under, comrade, affiliated with) and events (origin, participated in, pilots, killed, saved). The ontology is shared by all series and was not adapted to the evaluated work.

Extraction settings. The chapter pass reads spans of about 4,000 tokens with 500 tokens of overlap and records, for each scene, a summary, the characters who take part and up to four paragraphs that support the summary. The mention pass reads a window of four paragraphs before and after a mention, never across a hard scene break, with at most 40 windows per entity per volume, one window per call and at most 6,000 tokens per window (a denser cluster of mentions is split into consecutive windows). The long-context pass reads all of a character's paragraphs in a volume (up to 150,000 tokens) for gender, age and origin, and separately for background. A claim needs a self-reported confidence of at least 0.5 to be kept and 0.7 to fill a structured field. An evidence-adequacy gate rejects quotes shorter than six words, quotes in which the subject is not named within one paragraph, and quotes carrying irrealis markers ("would have", "might have").

Intervals. Single-valued attributes supersede: a value's interval closes when the next distinct value first appears. Relations in a declared conflict group supersede the same way; every other relation and multi-valued attribute is an open interval that opens at its earliest evidence, with its qualifier taken from evidence no later than that volume. Free-text values whose BGE-M3 embeddings have cosine similarity of at least 0.86 are clustered before intervals are built.

Generation. Every model call uses Gemini 3.6 Flash (Vertex AI, temperature 0.4, `thinking_budget` 0). The intended character-page template specifies appearance 2–4, personality 2–6, and history 3–12 sentences; shared-page summaries request one sentence. The archived NARRATIVEWIKI appearance prompt instead requests 3–4

Algorithm 1 Assembling the page of entity e at cutoff t.

Require: graph of verifier-accepted intervals; entity e; cutoff t
$I \leftarrow$ copies of $\{\, i \in \textsc{Intervals}(e) : i.start \le t \,\}$ ▷ a local view; the graph is not modified
if $I = \varnothing$ **then return** no page ▷ an entity with no interval by t has no page and is no link target
end if
for $i \in I$ **do**
  $i.quotes \leftarrow \{\, q \in i.quotes : \mathrm{vol}(q) \le t \,\}$ ▷ each quote filtered by its own volume
  if $i.end > t$ **then**
    $i.end \leftarrow \bot$ ▷ do not reveal that, or when, it ends
  end if
end for
$I \leftarrow \{\, i \in I : \textsc{Verified}(i) \,\}$ ▷ facts the verifier rejected are withheld
$\mathit{fields} \leftarrow \textsc{Render}(I)$ ▷ deterministic: names, attributes, relations, history
$\mathit{prose} \leftarrow \textsc{Generate}(I)$ ▷ model sees only I; no evidence ⇒ no call, empty section
$\mathit{page} \leftarrow \textsc{Link}(\mathit{fields}, \mathit{prose}, \{\, s : \mathrm{vol}(s) \le t \,\})$ ▷ surface forms and targets revealed by t
return *page* with cutoff-t title and navigation

sentences (Appendix Q). These are prompt requirements, not a measurement of output-length enforcement. Generation prompts request keys identifying the input lines supporting the text, which map back to source paragraphs. The parser also normalizes returned paragraph IDs during citation resolution, as in the codex example below. A shared-page summary is retained only if a second call judges every statement to be established by its input passages. A rejected summary is regenerated once and omitted if it fails again. Responses are cached by provider, model, prompt, and parameters, so rebuilding an unchanged configuration requires no new calls. All prompts are released verbatim with the code. Prompts for NARRATIVEWIKI and text-reading comparison systems omit the work's title and author; the closed-book diagnostic receives the title by design. A character with no STATUS claim is printed as *alive (assumed)*: a configured default with no evidence behind it, marked as assumed on the page, since the absence of a death claim does not establish that a character is alive.

Prompt and response defects. The extraction prompt of the reported runs exposes internal revision labels, module paths, and implementation notes in the STATUS and APPEARANCE descriptions. The prompt builder passes the same canonicalization note to the verifier as the *definition* of APPEARANCE, which the verifier is told to judge against; a rule for deduplicating values does not define a physical characteristic. The exhibited call accepts its appearance claim; the effect on other decisions is not measured. The codex response returns a paragraph ID where the prompt requests an input-line key; the parser maps it to its paragraph during citation resolution, without checking that the paragraph supports the sentence. All reported results use this configuration; a repaired prompt builder would require new runs, which this paper does not report.

Changes between builds. The reported build differs from the first build (Section 6.4, Table 21) by one batch of changes, made after the first build had been labeled and applied to every series alike. *Entity resolution:* a bare name and a fuller name first seen in a later volume go to the model instead of merging for free; a first name shared by several people is assigned per volume; fuzzy matching may not exchange a whole word; the cap on model-judged name pairs rose from 1,000 to 5,000; hyphenated names and titles of the form "X of the Y" are mined whole. *Ingestion:* publisher's back matter is dropped. *Scenes:* the participants of a scene are those who take part, not those mentioned, and each summary names up to four supporting paragraphs. *Citations:* generated text names the input lines it rests on, and scene lines print their paragraphs on timeline, pair and codex pages; the first build cited a chapter for a scene line and nothing for most prose. *Gates:* a prose answer that says it found nothing is discarded, and summaries on shared pages pass the check described above. *Windows:* the 6,000-token ceiling on a mention window, which the first build did not enforce on a dense cluster. One side effect: the first build held two entities for several main characters, and each entity received its own windows, so the same text was read twice; with the entities merged it is read once, and the claims stored for two such characters fell from 162 and 66 to 118 and 46 on as many windows. We did not isolate this change's contribution to the reported build's lower recall.

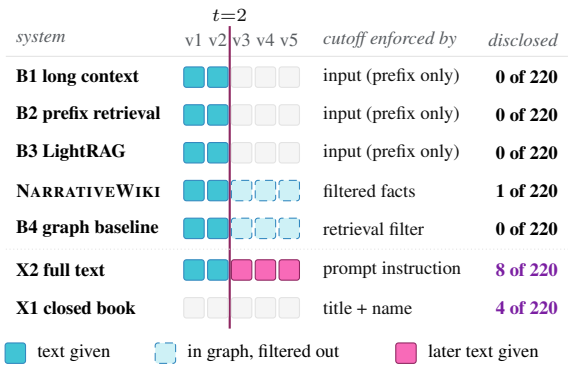


Figure 6: Information available to each system at cutoff 2 and its cutoff mechanism (left). Confirmed disclosures of assessed later facts on the original text, summed over cutoffs one to four (right). X1 and X2 are diagnostics.

E Comparison Systems

All comparison systems (B1–B4, X1, X2) use the same cast list, character-page template, generator, and intended character-page budget: the shared sections and sentence limits (appearance 2–4, personality 2–6, history 3–12), at most five quotes, and a shared citation format (Table 8). For systems supplied with passages, citations to paragraphs outside their input are removed, and characters absent from the eligible text receive no page but remain in the evaluation denominators. Settings were chosen on the development novel (*Middlemarch*) and fixed before the Anne runs. NarrativeWiki itself is not given the cast: it discovers entities and also publishes pages for other characters, relationships, codex entries, and a timeline. The shared template therefore controls the evaluated cast's character pages, while the complete artifacts differ in scope (Table 15). The exhibited comparison prompts request 2–4 appearance sentences; the archived NarrativeWiki appearance prompt requests 3–4 (Appendix Q). The template budget therefore does not imply identical prompt wording or verified enforcement of output length.

F Results at Every Cutoff

Tables 9 (original text) and 10 (treated text) report the individual cells underlying the means in Table 1; Figure 7 plots them. Long-context recall is approximately stable across cutoffs. NarrativeWiki has its highest recall on the first book, where it matches the closed-book diagnostic, and its lowest at $t = 2$. Closed-book recall declines from the first book to the fifth. LightRAG's low recall at $t = 1$ accompanies retrieved context concentrated on the opening chapters. With only 40 sampled atoms per cell, the precision results do not establish a trend across cutoffs. Every reported model call used the shared generator. Table 11 gives the mean, standard deviation, median, minimum, and maximum across cutoffs; Table 12 reports the same summaries for the paired differences discussed in Section 6.

	Adaptation and settings
B1	Every paragraph of volumes $1, \ldots, t$, labeled with its identifier, precedes the task so that pages at a cutoff share a cacheable prefix. Prompts exceeding the model's context limit would be marked unsupported rather than truncated; none exceeded it.
B2	Prefix-only index of consecutive paragraphs within a chapter (150 words). Okapi BM25 (k_1=1.5, b=0.75) and BGE-M3 cosine rankings for the character name alone and with seven facet words (appearance, personality, family, relations, goals, past, speech), fused by reciprocal rank (k=60); chunks added in fused order up to 20,000 tokens of retrieved text, then restored to reading order. Budgets count retrieved text only, estimated as characters divided by four, so a call's provider-counted input, which adds the prompt, can exceed them. Chunk size (150 or 300 words) and budget (6,000, 12,000 or 20,000 tokens) were chosen at t=2 of the development novel.
B3	LightRAG 1.5.7, default chunking (1,200 tokens) and its default extraction prompt and limits (at most 100 rows and 40 entities per chunk), mode `mix`, context capped at the same 20,000 tokens, reranking off. The index at cutoff t is the index at $t-1$ plus volume t, so no index sees a later volume. LightRAG supplies retrieval only: its retrieved entities, relations and chunks are passed to the shared generator, and its own extraction calls use the same model.
B4	The NarrativeWiki graph built from all five volumes. For each character, the facts visible at t and their evidence paragraphs of volumes $\leq t$, up to the same budget, are passed to the shared generator in place of NarrativeWiki's structured rendering (graph `anne@v2`).
X1	The shared prompt with the work's title, the character's name and no passages, with an instruction to write what is known of the character up to volume t; citations are empty by construction.
X2	Every paragraph of volumes 1–5 regardless of t, with the instruction to write for a reader of volume t and reveal nothing later.

Table 8: Adaptations of the comparison systems.

G Additional Results

Coverage of the evaluation. Table 13 lists the available builds and labels. B4's aggregate results are included in Table 1, with per-cutoff values below. The first build of NarrativeWiki and its prefix variants support the analyses in Sections 6.4 and 6.5.

What the disclosure search retrieves. Disclosure judgments are limited to sentences returned

t	System	Recall (M1)	Precision (M3)	Citation sufficiency (M8)	Disclosed (M4)	Assertions
1	B1	0.870 (20/23)	0.950 (38/40)	0.800 (32/40)	0/88	837
	B2	**0.913 (21/23)**	0.925 (37/40)	**0.850 (34/40)**	0/88	848
	B3	0.304 (7/23)	**0.975 (39/40)**	0.575 (23/40)	0/88	622
	B4	0.826 (19/23)	0.975 (39/40)	0.925 (37/40)	0/88	784
	NARRATIVEWIKI	0.826 (19/23)	0.825 (33/40)	0.575 (23/40)	0/88	**4,606**
	X1	0.826 (19/23)	0.725 (29/40)	–	**1/88**	807
	X2	0.870 (20/23)	0.900 (36/40)	0.700 (28/40)	0/88	862
2	B1	**0.870 (40/46)**	0.900 (36/40)	0.650 (26/40)	0/65	1,350
	B2	0.783 (36/46)	**1.000 (40/40)**	0.650 (26/40)	0/65	1,369
	B3	0.630 (29/46)	0.950 (38/40)	**0.800 (32/40)**	0/65	1,102
	B4	0.696 (32/46)	0.975 (39/40)	0.875 (35/40)	0/65	1,284
	NARRATIVEWIKI	0.761 (35/46)	0.718 (28/39)	0.487 (19/39)	0/65	**9,695**
	X1	0.652 (30/46)	0.875 (35/40)	–	**1/65**	1,280
	X2	0.761 (35/46)	0.900 (36/40)	0.675 (27/40)	**6/65**	1,432
3	B1	**0.818 (54/66)**	**0.950 (38/40)**	0.700 (28/40)	0/45	1,823
	B2	0.758 (50/66)	**0.950 (38/40)**	**0.800 (32/40)**	0/45	1,730
	B3	0.545 (36/66)	**0.950 (38/40)**	0.775 (31/40)	0/45	1,331
	B4	0.758 (50/66)	0.950 (38/40)	0.950 (38/40)	0/45	1,704
	NARRATIVEWIKI	0.803 (53/66)	0.875 (35/40)	0.500 (20/40)	**1/45**	**13,947**
	X1	0.652 (43/66)	0.897 (35/39)	–	0/45	1,511
	X2	0.803 (53/66)	0.825 (33/40)	0.575 (23/40)	0/45	1,740
4	B1	0.843 (75/89)	0.897 (35/39)	0.590 (23/39)	0/22	2,345
	B2	**0.854 (76/89)**	0.949 (37/39)	0.718 (28/39)	0/22	2,187
	B3	0.607 (54/89)	**0.950 (38/40)**	**0.775 (31/40)**	0/22	1,837
	B4	0.787 (70/89)	0.950 (38/40)	0.900 (36/40)	0/22	2,227
	NARRATIVEWIKI	0.820 (73/89)	0.850 (34/40)	0.500 (20/40)	0/22	**17,853**
	X1	0.663 (59/89)	0.800 (32/40)	–	**2/22**	2,016
	X2	0.843 (75/89)	0.950 (38/40)	0.625 (25/40)	**2/22**	2,421
5	B1	0.838 (93/111)	0.950 (38/40)	0.550 (22/40)	–	3,234
	B2	**0.865 (96/111)**	**0.975 (39/40)**	0.850 (34/40)	–	2,892
	B3	0.595 (66/111)	0.925 (37/40)	**0.900 (36/40)**	–	2,440
	B4	0.739 (82/111)	0.975 (39/40)	0.925 (37/40)	–	2,981
	NARRATIVEWIKI	0.802 (89/111)	0.675 (27/40)	0.500 (20/40)	–	**23,613**
	X1	0.622 (69/111)	0.775 (31/40)	–	–	2,570
	X2	0.793 (88/111)	0.975 (39/40)	0.550 (22/40)	–	3,196

Table 9: Results at every cutoff. Rates are followed by their counts. Disclosure is the number of assessed later facts stated or implied over $|F_t|$ and is undefined at $t = 5$; citation sufficiency has no value for X1, which cites nothing; assertions is the size of the cutoff's assertion inventory, counting a statement once for every page that prints it. A precision denominator below 40 reflects excluded frame errors. B4 (the graph baseline, which reuses NARRATIVEWIKI's graph) is unhighlighted; the treated-text cells are in Table 10. Bold: highest rates and assertion count at each cutoff among B1, B2, B3 and B5, and each non-zero disclosure count. Cyan, magenta, and violet highlight performance maxima, disclosures, and assertion counts, respectively.

by candidate search. We therefore test retrieval on seeded sentences with known disclosures (Table 14). Search retrieves every verbatim claim, about 85% of named paraphrases and about 70% of named implications; pronouns on another character's page reduce retrieval to roughly one third. The corresponding rates are 220 of 220 for verbatim claims, 0.845 (186/220) and 0.768 (169/220) for paraphrases, and 0.714 (157/220) and 0.623 (137/220) for implications. Search produced 5,702 candidate sentences for NARRATIVEWIKI, 520 for B1 and 12,689 across systems and builds. A lexical-screen hit on X2 exposed an unmatched phrase ("to marry"); after correcting the matcher, 144 sentences were reassessed. Of 224 eligible claims passed through the same search, 14 became candidates for a later fact about the same character. Retrieval alone does not count as disclosure.

The annotator model wrote and checked each fixture for its subject, event and object, independently of whether search retrieved it. An earlier fixture version, in which many implications did not establish their target fact, had retrieval of approximately one quarter and is released alongside the validated version. No second reader validated the fixtures. The search-independent checks of Section 6.2 also have limits: lexical screening detects only new names, while zero later-volume assertions

Metric	t	B1	B2	B3	B4	B5	X1	X2
Recall	1	**0.957 (22/23)**	0.870 (20/23)	0.348 (8/23)	0.652 (15/23)	0.870 (20/23)	0.000 (0/23)	0.826 (19/23)
	2	**0.891 (41/46)**	0.783 (36/46)	0.457 (21/46)	0.674 (31/46)	0.826 (38/46)	0.000 (0/46)	0.696 (32/46)
	3	0.818 (54/66)	0.727 (48/66)	0.500 (33/66)	0.758 (50/66)	**0.848 (56/66)**	0.015 (1/66)	0.667 (44/66)
	4	0.764 (68/89)	0.775 (69/89)	0.539 (48/89)	0.663 (59/89)	**0.843 (75/89)**	0.000 (0/89)	0.764 (68/89)
	5	0.793 (88/111)	0.766 (85/111)	0.505 (56/111)	0.775 (86/111)	**0.829 (92/111)**	0.000 (0/111)	0.766 (85/111)
Precision	1	**0.975 (39/40)**	**0.975 (39/40)**	0.925 (37/40)	0.925 (37/40)	0.750 (30/40)	0.200 (8/40)	0.825 (33/40)
	2	0.950 (38/40)	**0.975 (39/40)**	0.949 (37/39)	**0.975 (39/40)**	0.700 (28/40)	0.150 (6/40)	0.825 (33/40)
	3	0.900 (36/40)	0.925 (37/40)	**1.000 (40/40)**	0.975 (39/40)	0.875 (35/40)	0.125 (5/40)	0.825 (33/40)
	4	0.825 (33/40)	0.950 (38/40)	**1.000 (40/40)**	0.923 (36/39)	0.750 (30/40)	0.150 (6/40)	0.825 (33/40)
	5	**0.950 (38/40)**	0.925 (37/40)	0.925 (37/40)	0.900 (36/40)	0.775 (31/40)	0.100 (4/40)	0.925 (37/40)
Citation suff.	1	0.875 (35/40)	**0.925 (37/40)**	0.625 (25/40)	0.850 (34/40)	0.500 (20/40)	–	0.575 (23/40)
	2	0.675 (27/40)	**0.900 (36/40)**	0.769 (30/39)	**0.900 (36/40)**	0.700 (28/40)	–	0.550 (22/40)
	3	0.425 (17/40)	0.825 (33/40)	**0.875 (35/40)**	**0.875 (35/40)**	0.575 (23/40)	–	0.475 (19/40)
	4	0.300 (12/40)	0.925 (37/40)	0.800 (32/40)	**0.949 (37/39)**	0.700 (28/40)	–	0.400 (16/40)
	5	0.375 (15/40)	0.850 (34/40)	**0.900 (36/40)**	**0.900 (36/40)**	0.650 (26/40)	–	0.375 (15/40)
Disclosed	1	0/88	0/88	0/88	0/88	0/88	0/88	**8/88**
	2	0/65	0/65	0/65	**2/65**	**2/65**	0/65	**8/65**
	3	0/45	0/45	0/45	0/45	**2/45**	0/45	**6/45**
	4	0/22	0/22	0/22	0/22	0/22	0/22	**1/22**
	5	–	–	–	–	–	–	–

Table 10: Every cell of the treated text behind Table 1: recall over the eligible gold facts, precision and citation sufficiency over 40 sampled statements, and stated or implied disclosures over the later gold facts (none are left to disclose at $t = 5$). The closed-book generator cites nothing. Bold cyan cells mark the highest rates among B1–B5 at each cutoff; bold magenta cells mark non-zero disclosure counts.

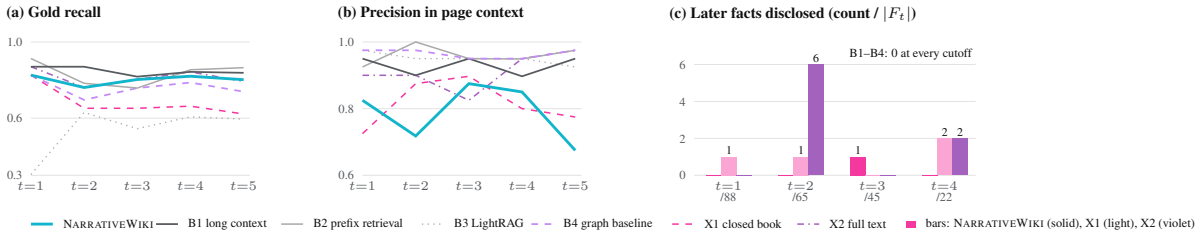


Figure 7: Recall and precision at each cutoff (lines), and counts of assessed later facts disclosed (bars), original text. Dashed lines denote the X1 and X2 diagnostics; NarrativeWiki is cyan. B1–B4 have zero observed disclosures at every cutoff. Disclosure denominators are $|F_t| = 88, 65, 45, 22$ at $t = 1, \ldots, 4$; disclosure is undefined at $t = 5$.

in a sample of approximately 160 atoms gives limited evidence about rare disclosures. Overlapping cutoff wikis and clustering within pages reduce the effective sample size.

Matched and whole-artifact populations. Table 15 separates sampled assertions by page type. The comparison systems publish only cast character pages; on this matched population, 31 of NarrativeWiki's 35 sampled atoms are supported. This small subsample does not support a precise comparison with the baselines, and no equivalence test is reported. Of NarrativeWiki's 42 page-context negatives, 30 occur on relationship pages; 27 of those 30 are supported as statements but placed under a pair that shares no episode of the scene.

For every NarrativeWiki page type, observed citation sufficiency is below the overall values of B1 and B2. Cited paragraphs establish 14 of 35 atoms on cast pages, 60 of 102 on relationship pages and 11 of 22 on the timeline. Timeline lines can carry up to four paragraphs selected to support a chapter-level summary.

The graph baseline (B4). Table 9 reports B4 at every cutoff, using the reported NarrativeWiki graph. Mean recall is 0.761, versus 0.802 for NarrativeWiki: lower at four cutoffs and equal at one. Mean precision is 0.965. No assessed later fact was disclosed (0 of 220); B4 publishes 1,701 distinct assertions per cutoff and has 1 lexical-screen hit, the same expanded page title found in the prefix baselines.

B4 generates cast character pages from a budgeted selection of retrieved graph facts; Narra-

Recall ($n = 5$)

Statistic	B1	B2	B3	B4	B5	X1	X2
Mean	**0.848**	0.834	0.536	0.761	0.802	0.683	0.814
SD	0.022	0.063	0.133	0.049	0.026	0.082	0.043
Median	0.843	0.854	0.595	0.758	0.803	0.652	0.803
Min	0.818	0.758	0.304	0.696	0.761	0.622	0.761
Max	0.870	0.913	0.630	0.826	0.826	0.826	0.870

Precision ($n = 5$)

Statistic	B1	B2	B3	B4	B5	X1	X2
Mean	0.929	0.960	0.950	**0.965**	0.789	0.814	0.910
SD	0.028	0.029	0.018	0.014	0.087	0.071	0.058
Median	0.950	0.950	0.950	0.975	0.825	0.800	0.900
Min	0.897	0.925	0.925	0.950	0.675	0.725	0.825
Max	0.950	1.000	0.975	0.975	0.875	0.897	0.975

Citation sufficiency ($n = 5$)

Statistic	B1	B2	B3	B4	B5	X1	X2
Mean	0.658	0.774	0.765	**0.915**	0.512	–	0.625
SD	0.098	0.088	0.118	0.029	0.035	–	0.064
Median	0.650	0.800	0.775	0.925	0.500	–	0.625
Min	0.550	0.650	0.575	0.875	0.487	–	0.550
Max	0.800	0.850	0.900	0.950	0.575	–	0.700

Disclosure rate ($n = 4$)

Statistic	B1	B2	B3	B4	B5	X1	X2
Mean	0.000	0.000	0.000	0.000	**0.006**	**0.029**	**0.046**
SD	0.000	0.000	0.000	0.000	0.011	0.042	0.053
Median	0.000	0.000	0.000	0.000	0.000	0.013	0.045
Min	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Max	0.000	0.000	0.000	0.000	0.022	0.091	0.092

Assertions ($n = 5$)

Statistic	B1	B2	B3	B4	B5	X1	X2
Mean	1,918	1,805	1,466	1,796	**13,943**	1,637	1,930
SD	924	781	698	849	7,315	679	904
Median	1,823	1,730	1,331	1,704	13,947	1,511	1,740
Min	837	848	622	784	4,606	807	862
Max	3,234	2,892	2,440	2,981	23,613	2,570	3,196

Table 11: Distribution over cutoffs of every per-cutoff cell: unweighted mean, sample standard deviation, median, lowest and highest value over the n cutoffs measured ($t = 1..5$; disclosure $t = 1..4$). Disclosure rate is stated or implied disclosures over $|F_t|$. Rate means equal the point estimates of Table 1; the assertion panel counts every assertion as it occurs on a page, not the distinct assertions of Table 1. Bold: highest performance and assertion-count means among B1–B5, and each non-zero mean disclosure rate. Cyan highlights performance maxima, magenta non-zero disclosure means, and violet assertion count.

Metric	B5−	Mean	SD	Median	Min	Max
Recall	B1	**−0.045**	0.037	−0.036	−0.109	−0.015
	B2	−0.032	0.050	−0.034	−0.087	0.045
	B3	**0.266**	0.150	0.213	0.130	0.522
	X1	0.120	0.072	0.152	0.000	0.180
	X2	−0.011	0.021	0.000	−0.043	0.009
Disclosure	X1	−0.024	0.048	−0.013	−0.091	0.022
	X2	**−0.040**	0.060	−0.045	−0.092	0.022

Table 12: Paired differences of NARRATIVEWIKI (B5) from each system, mean over cutoffs of the per-cutoff difference on the entities both systems have; SD, median, lowest and highest value are over the per-cutoff differences (five cutoffs for recall, four for disclosure). Recall minus X1 compares source-conditioned output with the closed-book diagnostic. Bold cyan cells mark the largest mean difference in each direction.

	Cutoffs built	M1, M3, M4	M8
B1, B2, NARRATIVEWIKI	1–5	labeled	labeled
on prefix indexes	1–4	labeled	labeled
B3, B4, X2	1–5	labeled	labeled
X1 (cites nothing)	1–5	labeled	–
NARRATIVEWIKI, first build	1–5	labeled	labeled
from a prefix	2, 3	labeled	–

Table 13: What was built and labeled. Every label is by an LLM annotator (Section 5).

TIVEWIKI renders all eligible facts and also publishes relationship, codex and timeline pages. B4's higher precision, together with the 30 relationship-page negatives among NARRATIVEWIKI's 42 page-context negatives, suggests that publication context contributes to the gap in Table 1. This comparison does not isolate context from selection and generation.

B4's cited paragraphs suffice for 0.915 of sampled assertions, and none of its 200 sampled citations is judged wholly unrelated to the assertion. The corresponding whole-artifact value for NARRATIVEWIKI is 0.512; on character pages it is 29 of 64, versus 183 of 200 for B4. These observations show that evidence stored in the common graph can support generated assertions, while NARRATIVEWIKI's evidence selection or attachment is less reliable.

Identity dependence. Table 17 reports the measure defined in Section 6.5; its mean is 0.217. Every prefix-built character has a counterpart in the

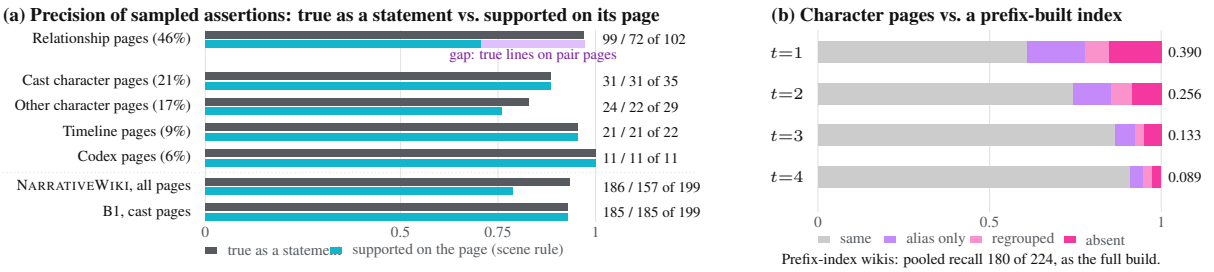


Figure 8: Context collapse (RQ4) and identity dependence (RQ5), original text. (a) On relationship pages nearly every line is true, but a quarter sit under a pair that shares no episode; on cast pages the two measures agree. (b) NarrativeWiki's character pages against an entity index discovered from volumes $1..t$; the printed rate is the share that differ.

Seeded sentence	Retrieved	Rate
Gold claim verbatim	220	1.000
Paraphrase, names the character	186	0.845
Paraphrase, pronoun, own page	169	0.768
Paraphrase, pronoun, other page	68	**0.309**
Implying, names the character	157	0.714
Implying, pronoun, own page	137	0.623
Implying, pronoun, other page	78	**0.355**

Table 14: Recall of the disclosure candidate search on seeded disclosures: 220 fact-by-cutoff units ($t = 1..4$), one paraphrase and one implying sentence per later fact, each checked to establish its fact. Bold magenta cells mark the lowest rate of each kind.

	Share	n	Page	Atom	Suff.
Character pages, cast	0.206	35	31	31	14
Character pages, other	0.171	29	22	24	15
Relationship pages	**0.463**	102	**72**	**99**	60
Codex pages	0.057	11	11	11	2
Timeline	0.090	22	21	21	11
Index	0.012	0	–	–	–
NarrativeWiki, all pages	1.000	199	157	186	102
B1 (cast pages only)	1.000	199	185	185	131
B2 (cast pages only)	1.000	199	191	191	154

Table 15: The precision samples by kind of page, pooled over cutoffs. Share: of the system's assertions on that kind of page; n: sampled atoms; Page and Atom: supported on the page that prints them (relationship pages under the pair rule of Appendix B), and as statements; Suff.: established by their own cited paragraphs. Bold lavender cells highlight the largest kind of page and its gap between assertions true as statements and assertions supported on the page.

full build, although counterpart existence does not imply identical alias grouping.

Some differences arise deterministically during candidate mining. Of the full build's candidates first occurring by t, 109, 111, 78 and 44 are absent from prefix mining ($t = 1..4$). Most, 91, 87, 60 and 32, occur fewer than three times in the prefix and reach the miner's frequency threshold only with later occurrences. The remaining 18, 24, 18 and 12 fail a separate rule: a single capitalized word must stand alone in at least three quarters of its occurrences in some volume. Bare *Gilbert* appears alone in 62 of 113 book-one occurrences; the other 51 are in *Gilbert Blythe*. Prefix mining therefore omits the alias, while later volumes allow the full build to retain it. Disabling this rule admits 352 additional full-build candidates, mostly sentence-initial words. We retain the reported rule because changing it would alter the evaluated index.

Pages absent from the prefix index at $t = 1$ include `walter`, `jerry` and `marshall`, discussed below, and Anne's mother, whose full name occurs once in book one. The four prefix indexes use 2.61M input and 0.26M output tokens, including 0.37M and 0.04M in calls absent from the full build's cache. The measure captures entity-index sensitivity to construction scope; wiki outcomes on these indexes appear in Table 18.

Wikis built on the prefix indexes. Table 18 evaluates the complete reported pipeline on each prefix $t = 1..4$, using that prefix's entity index. Pooled recall is 180 of 224 gold-fact assessments for the prefix builds and 180 of 224 for the full build. Per-cutoff differences comprise one to three facts in either direction: prefix builds lose a minor character's trait and gain an office or prize omitted by the full build. Pooled precision is 134 of 159, versus 130 of 159, on different samples. Citation sufficiency is 91 of 159, versus 82 of 159, using different samples with labels finalized by Claude.

Neither build states a later fact. Both contain the same implied disclosure at $t = 3$, the bridesmaid relation inverted from eligible text. Prefix builds publish fewer character pages at early cutoffs: `walter`

	$t = 2$	$t = 3$
Pages, full build	133	173
Pages, prefix build	107	151
In both builds	100	145
same title	100	145
same alias list	80	128
alias not in $D_{\leq t}$ (full)	0	0
In one build only	40	34
later-anchored	**7**	**9**
second page, same person	9	9
same person, new address	9	5
minor entity	15	11
stating a later fact	0	0
Recall, prefix build	0.804 (37/46)	0.848 (56/66)
Recall, full build	**0.848 (39/46)**	**0.864 (57/66)**
Precision, prefix build	**0.769 (30/39)**	**0.718 (28/39)**
Precision, full build	0.718 (28/39)	0.692 (27/39)
Later facts disclosed	0/65	0/45
Prefix build tokens (M)	12.03 / 0.68	17.94 / 1.01

Table 16: Construction scope (character pages), first build of NARRATIVEWIKI: built from volumes $1..t$ alone (prefix) against the same cutoff of its five-volume build (full; 33.14M input and 1.81M output tokens once, for all cutoffs). Precision is under the pair rule of Appendix B. Disclosure counts are identical in the two builds; tokens are input / output, in millions. Bold magenta cells mark later-anchored pages; bold cyan cells mark the higher of the two builds on recall and precision.

and `jerry`, retained by the full build at every cutoff because of book-five characters, never appear in prefix builds; `marshall`, from book four, appears only at $t = 4$. Prefix-specific discovery avoids using later text in entity construction while retaining the observed pooled gold recall. This is not evidence of equivalence on all outcomes: the approach requires a build per cutoff, and the four prefix builds together use more tokens than the single full build.

Construction scope. Table 16 compares prefix and full graphs made with first-build code; this complete graph comparison was not repeated for the reported build. The pooled recall difference is -0.027 (93 vs 96 of 112), with seven discordant facts; the precision difference is $+0.038$ (58 vs 55 of 78), using separate samples. These comparisons cannot isolate construction-scope effects from differences in entity indexes and model runs.

At $t = 2$, later-anchored pages include `walter` (Walter Blythe of book five, 127 of the entity's 130 mentions, but an early page describing Anne's father); `di` (Di Blythe, with an early duplicate page for Diana Barry under a nickname used once in the first two books); `mary` (Mary Vance, showing the twins' mother); `jerry` (Jerry Meredith, merging a book-one hired boy with a book-two agent); and `marshall`, `baxter` and `sam`. An entity-merge audit flags 42 entities in the full build and 2 in a volume-one build.

Shared pages have identical titles and no full-build alias absent from $D_{\leq t}$. Relation lines differ in both directions: at $t = 2$, 54 occur in both builds, 93 only in the full build and 68 only in the prefix build. Separate runs and entity lists prevent attributing these differences solely to later volumes. Prose on shared pages was not compared.

Later-anchored entities in the reported build. The reported build adjudicates bare names against fuller forms first appearing later, and assigns shared first names volume by volume to the locally dominant full name. Its `walter` entity now contains only Anne's father (three mentions), with Walter Blythe appearing separately from book five.

A separate screen flags character-page entities with at least ten mentions and at least four fifths occurring after the cutoff: 6 of 117 at $t = 2$ and 3 of 158 at $t = 3$. At $t = 2$, these are `susan` (3 of 204 mentions by book two; an empty page dominated by Susan Baker of books four and five); `sam` (2 of 10, with prose saying "Sam does not appear in the story provided"); `joseph` (5 of 54, an early pupil's uncle under an entity later dominated by others); `jerry-corcoran` (6 of 120, correctly described at $t = 2$ but later absorbing Jerry Meredith); and `stella-maynard` and `billy-andrews`, correctly resolved people introduced early. This screen identifies entities dominated by later mentions. It neither establishes dependence for every flagged page nor determines what a prefix build would publish instead.

Cost. Table 19 breaks down model use by stage; extraction dominates. Prose and shared-page summaries are generated once per distinct filtered fact set, across all five cutoffs, and included in the total. Verification includes rechecks after one re-extraction, so its count slightly exceeds a cold build from an empty cache. The first build used 33.14M input and 1.81M output tokens under the same accounting. B4 adds 1.11 / 0.36 million tokens (input / output) of page generation to the graph it shares with NARRATIVEWIKI; latency, hardware and the local evaluation models are not costed for any system.

Long-context input grows with the prefix (2.2M, 7.0M, 10.8M, 19.6M and 29.7M tokens at $t =$

t	Pages	Same	Alias	Regrouped	Absent	Identity dependence	Regrouped or absent	Mentions
1	59	36	10	4	9	**0.390 (23/59)**	**0.220 (13/59)**	**0.943**
2	117	87	13	7	10	0.256 (30/117)	0.145 (17/117)	0.967
3	158	137	9	4	8	0.133 (21/158)	0.076 (12/158)	0.982
4	191	174	7	5	5	0.089 (17/191)	0.052 (10/191)	0.994

Table 17: Identity dependence of the reported NARRATIVEWIKI on the original text. Each character page of the five-volume build at cutoff t is compared with the entity index built from volumes $1..t$ alone. *Same*: a prefix-built character has exactly its set of names seen by t. *Alias*: one counterpart, with a name linked in one build only. *Regrouped*: names that one build gives one character the other gives two. *Absent*: no prefix-built entity carries any of its names. *Mentions*: character mentions in volumes $1..t$ under the prefix index, as a share of the full build's. One run per index, so model variance is inside the differences. Bold magenta cells mark the cutoff where identity dependence is largest.

	Build	$t = 1$	$t = 2$	$t = 3$	$t = 4$
Character pages	full	59	117	158	191
	prefix	48	108	155	185
Recall	full	0.826 (19/23)	0.761 (35/46)	0.803 (53/66)	**0.820 (73/89)**
	prefix	0.826 (19/23)	0.761 (35/46)	**0.818 (54/66)**	0.809 (72/89)
Precision	full	0.825 (33/40)	0.718 (28/39)	0.875 (35/40)	**0.850 (34/40)**
	prefix	**0.950 (38/40)**	**0.750 (30/40)**	0.875 (35/40)	0.795 (31/39)
Citation sufficiency	full	0.575 (23/40)	**0.487 (19/39)**	0.500 (20/40)	0.500 (20/40)
	prefix	**0.650 (26/40)**	0.425 (17/40)	**0.600 (24/40)**	**0.615 (24/39)**
Later facts disclosed	full	0/88	0/65	**1/45**	0/22
	prefix	0/88	0/65	**1/45**	0/22
Assertions	full	4,606	9,695	13,947	17,853
	prefix	3,977	8,732	13,432	17,723
Tokens, in / out (M)	prefix	5.68 / 0.31	11.95 / 0.65	18.04 / 0.99	24.04 / 1.37

Table 18: The reported NARRATIVEWIKI built on entity indexes discovered from the prefix (*prefix*: one build per cutoff, every stage on volumes $1..t$) against the same cutoff of its five-volume build (*full*). Metrics as in Table 9; precision is on a different sample of 40 assertions in each build. The prefix builds received a single LLM labeling pass; the full build's labels were finalized by Claude Opus 5.5 (Appendix B). The full build's tokens, for all five cutoffs at once, are in Table 19. Bold cyan cells mark the higher rate where the builds differ; bold magenta cells mark non-zero disclosure counts.

1..5), which is sent once per page. The all-cutoff NARRATIVEWIKI build uses 33.1M input tokens. LightRAG's 5.08 / 1.01 million tokens (input / output) comprise 2.36M input and 0.72M output tokens for five incremental indexes and 2.72M and 0.29M for page generation. Prefix retrieval indexes use local embeddings and BM25, so its 3.42 / 0.37 million tokens cover page generation. The closed-book diagnostic uses 0.06 / 0.15.

Verification. At audit, the verifier had marked 21.2% of 2,535 facts unsupported by their cited evidence; these facts are withheld. Table 19 counts verification calls rather than distinct facts. On a challenge set of 81 items created by minimal substitutions in source-verified facts, the verifier accepted 1 of 46 labeled unsupported candidates and rejected 0 of 35 labeled supported ones (Appendix K). This synthetic challenge does not establish performance on naturally occurring extraction errors.

Facts NARRATIVEWIKI misses. At $t = 5$, NARRATIVEWIKI misses 22 of the 111 gold facts; B1 or B2 conveys 18 of those omissions. Examples include Gilbert's teaching appointment at White Sands and Cooper Prize, Diana's role as improvement-society treasurer, Rilla's position as the youngest Blythe child, Leslie's mother's pressure to marry, Mr. Phillips's interest in Prissy Andrews, and Paul Irving's place of origin. Conversely, four facts are conveyed by NARRATIVEWIKI but neither B1 nor B2: Anne's fictional heroine, her mother's letters, Gilbert's boat rescue of Anne and Walter's hatred of a schoolmate. Four facts are conveyed by none of these three systems.

Typical errors. NARRATIVEWIKI still merges namesakes. Priscilla Grant's page lists the book-one schoolgirl "Prissy Andrews" as an alias. The `james-a` page joins Mr. Harrison of book two, who also has a separate page, with a deceased James of

Stage	Calls	Input	Output
Entity discovery	2,771	1.39	0.14
Chapter pass	452	2.74	**0.97**
Mention pass	**2,932**	**14.04**	0.43
Long-context pass	980	7.10	0.11
Arbitration	64	0.02	0.00
Verification	2,816	6.21	0.04
Prose	953	0.89	0.15
Shared-page summaries	386	0.73	0.02
Total, reported runs	11,354	33.13	1.86
Rebuild from cache	0	0	0

Table 19: Size of the reported NARRATIVEWIKI build for all five cutoffs, including verification rechecks after one re-extraction: model calls, and input and output tokens in millions. Every call of the build is counted, whether or not the provider served it at a discount, so the token column of Table 1 is comparable across systems. A rebuild from the call cache sends nothing. Ingestion, assembly and linking make no model call. Bold violet cells identify the stage consuming the most resources in each column.

book three and a book-four sailor. A duplicate page for Anne, `shirley`, also absorbs her son Shirley at $t = 5$ ("the little brown boy").

Other errors include a remark by Mrs. Barry attributed to Marilla, an inverted relationship (Section 6.2), intervals persisting beyond their validity and an exhibition prize labeled "Fought in". Generated commentary about the input can reach pages: one subject "does not appear in the story provided", and 5 of the 341 non-timeline pages at $t = 4$ retain source-line keys ("[R3, R4, S1, S4]"; Table 29). Two first-build errors were not observed in the reported build: Anne's mother's death specified as "of yellow fever", beyond the text, and a merger of two men named George.

B1 and X2 also misattribute biographies: at $t = 5$, both place one man's courtship and marriage on another man's page. The closed-book diagnostic imports material outside the five-book corpus, describing Anne's years at "Windy Willows".

Uncertain disclosure verdicts. System-level uncertain counts at $t = 1..4$ are 3 (B1), 0 (B2), 1 (B3), 0 (B4), 4 (B5), 2 (X1), 6 (X2). For NARRATIVEWIKI, four verdicts arise from one relation line between Anne and Gilbert, "Romantically involved with — harbors romantic feelings for her". The interval opens in book one from a friend's report that Gilbert kept a rose Anne dropped. Assessed at $t = 1$ and $t = 2$ against two later facts about the couple, the line yields four uncertain judgments because it could suggest their book-four marriage.

System	Original	Treated
B1	0/88	0/88
B2	0/88	0/88
B3	0/88	0/88
B4	0/88	2/88
B5 (NARRATIVEWIKI)	1/88	2/88
X1	3/88	0/88
X2	8/88	17/88

Table 20: Confirmed unique subject facts disclosed, over unique assessed subject facts. The unit is (character, claim, reveal volume): repeated cutoffs collapse, while reciprocal character facts remain distinct. These are not independent event or error counts.

Unique facts and error mechanisms. Table 20 collapses repeated fact–cutoff assessments to unique subject facts. On the original text, NARRATIVEWIKI's confirmed disclosure concerns one fact from one inverted-bridesmaid error. On the treated text, its four confirmed assessments cover two reciprocal marriage facts at two cutoffs, arising from a single prediction-to-marriage failure mechanism. B4's two treated assessments concern those same reciprocal facts. These witness-based descriptions do not establish that every output error has been found or that error mechanisms are independent. Grouping systems by information available to page generation includes B4 alongside B1–B3 and B5: together they have 1 of 1,100 original-text assessments, or 9 of 1,100 including uncertain verdicts.

The two builds and the pair rule. Table 21 compares the first and reported builds. The revision (Appendix D) lowered recall and raised citation sufficiency, merged two of the protagonist's three entities, removed three of four lexical-screen hits and introduced the implied disclosure. Precision is 0.751 and 0.789 under the main scene-sharing rule; 0.604 and 0.628 when the pair must participate in the line's own event; and 0.949 and 0.934 when assertions are assessed without page context. Rule choice changes the observed estimate more than the revision does. At 40 atoms per cutoff, these estimates do not establish equal precision across builds.

Two earlier estimates are superseded because their judgment rules differ: 0.822, allowing both pair members anywhere in a chapter, and 0.748, an initial reported-build pass. The provisional and finalized passes agree on 90 of 99 pair verdicts. Eight

	First	Reported
Recall	**0.855**	0.802
Precision, scene rule	0.751	**0.789**
Precision, line rule	0.604	**0.628**
True as statements	**0.949**	0.934
Citation sufficiency	0.112	**0.512**
With a paragraph citation (all)	0.198	**0.930**
Later facts disclosed	**0 of 220**	1 of 220
Lexical screen hits	4	**1**
Assertions per cutoff	20,823	13,943
distinct	8,808	6,970
Pair lines sharing an episode	76 of 116	**75 of 102**
Participants per scene record	9.0	5.9
Tokens, in / out (M)	33.14 / 1.81	33.13 / 1.86

Table 21: The two builds of NARRATIVEWIKI, means over cutoffs $t = 1..5$. Every relationship line of both samples was read by Claude Opus 5.5 under one rule; all other labels come from each build's own labeling round, the first build's before the revision and the reported build's after. Bold marks the better of the two builds where a direction exists; cyan emphasizes selected improvements, and magenta marks the observed disclosure.

		$t = 1$ (14)	2 (21)	3 (31)	4 (37)	5 (38)	Pooled (141)
original	B1	11	**19**	25	30	33	118
	B2	**12**	15	22	29	32	110
	B3	3	13	12	19	20	67
	B4	**12**	15	22	29	28	106
	NARRATIVEWIKI	**12**	18	**27**	**34**	**35**	**126**
	X2	12	16	25	31	31	115
treated	B1	**14**	**19**	23	27	29	112
	B2	12	14	19	23	26	94
	B3	5	9	12	16	14	56
	B4	9	11	21	21	24	86
	NARRATIVEWIKI	12	**19**	**29**	**35**	**36**	**131**
	X2	12	14	16	26	27	95

Character	Text	B5	B1	B2	B3
Anne (81)	original	**78**	64	60	41
	treated	**72**	61	53	41
Gilbert (39)	original	31	33	**34**	12
	treated	**38**	31	30	8
Diana (21)	original	17	**21**	16	14
	treated	**21**	20	11	7

Table 22: Gold facts conveyed about the three central characters (Anne Shirley, Gilbert Blythe, Diana Barry). Top: at each cutoff, eligible fact checks in parentheses; bold, best of B1–B5 per column and text. Bottom: per character, pooled over $t = 1..5$ (fact checks in parentheses); B5 is NARRATIVEWIKI; bold, best per row.

of nine disagreements involve lines accepted in the finalized pass; four occur on Anne's duplicate page, whose identity the provisional pass rejects. Substituting the provisional pair verdicts while retaining all other labels yields precision 0.753.

The page-context error has a structural mechanism: scene records summarize chapters (median 38 paragraphs) and list their participants; pair pages print every line of every record naming both members. The revision requested participants rather than merely mentioned characters, reducing participants per record from 9.0 to 5.9 and implied pairs from 10,216 to 4,604. The share of sampled pair lines whose characters share an episode rises from 76 of 116 to 75 of 102. Episode-level records are a possible remedy that has not been tested.

H Where Accumulation Helps

Section 6.3 reports three situations in which NARRATIVEWIKI is ahead of the systems that rewrite each page per cutoff. This appendix gives the data behind them. Nothing here required a model call: all values come from the stored recall labels and the built wikis. On the treated text, characters are named by their original names.

Central characters. The three central characters are those with the most gold fact checks: Anne Shirley (81 of 335), Gilbert Blythe (39) and Diana Barry (21). They are selected by that count alone, which does not depend on any system's output; the decision to report recall by this split was made after the results were known. Table 22 gives the per-cutoff counts. On treated text NARRATIVEWIKI's recall on them rises from 86% at $t = 1$ to 95% at $t = 4$ and $t = 5$, while long context falls from 100% to 73–76% and prefix retrieval from 86% to 61–68%. On the original text the lead comes from the protagonist alone; on the treated text NARRATIVEWIKI leads on all three characters (Table 22).

Why pages diverge in length. Comparison systems write one page per character in one call under a shared template with sentence limits per section, and the retrieval systems pass at most 20,000 tokens of passages (Appendix E). Their pages therefore stay roughly the same length however much the story says about a character. NARRATIVEWIKI renders every eligible claim. Table 23 gives the length of Anne Shirley's page and of each wiki, in words with citations removed. Length is not monotone: Anne Shirley's NARRATIVEWIKI page has 3,719 words at $t = 4$ and 3,439 at $t = 5$.

Facts kept across cutoffs. A comparison system writes each cutoff's page afresh, so a fact present at t can be absent at $t + 1$. For NARRATIVEWIKI, every claim eligible at t is still eligible at $t + 1$; a

Measure	System	$t=1$	$t=2$	$t=3$	$t=4$	$t=5$
Pages published	NARRATIVEWIKI	97	193	275	345	440
	B1, B2, B3	14	21	25	33	43
Words in the wiki	NARRATIVEWIKI	38,910	80,165	113,891	148,154	194,562
	B1	6,058	10,585	14,018	18,170	24,625
	B2	6,627	10,762	13,217	17,352	22,989
	B3	4,736	8,384	10,386	14,273	18,951
Words on Anne Shirley's page	NARRATIVEWIKI	1,813	2,363	2,688	3,719	3,439
	B1	864	852	913	984	1,157
	B2	699	553	708	711	688
	B3	620	731	653	554	410
Related characters on Anne Shirley's page	NARRATIVEWIKI	18	26	31	39	42
	B1	10	10	13	12	15
	B2	8	5	9	10	10
	B3	8	8	8	5	5

Table 23: Size of what each system publishes at each cutoff, original text. Words exclude front matter and citation blocks. NARRATIVEWIKI repeats a relation line on both characters' pages and on the pair's page, so its word count overstates distinct content; distinct assertions per cutoff (6,970 for NARRATIVEWIKI against 1,821, 1,718 and 1,404 for B1–B3) are the deduplicated measure: assertions identical after lower-casing and folding punctuation count once, so paraphrases, reciprocal phrasings and alias variants still count separately. Related characters: relationship blocks on the page, each linked to that character's page.

	Central three		Other 40		Facts lost	
System	orig.	treat.	orig.	treat.	orig.	treat.
B1	0.837	0.794	0.845	0.830	10	14
B2	0.780	0.667	**0.871**	**0.845**	4	24
B3	0.475	0.397	0.644	0.567	19	13
B4	0.752	0.610	0.758	0.799	12	14
B5	**0.894**	**0.929**	0.737	0.773	**1**	**1**
X2	0.816	0.674	0.804	0.789	7	11

Table 24: Where accumulation helps. Recall pooled over $t = 1..5$ on the three characters with the most gold assessments (Anne, Gilbert, Diana: 141 of 335) and on the other 40 (194), on original and treated text. Facts lost: gold facts conveyed at t and not at $t+1$, summed over $t = 1..4$ (lower is better). B5 is NARRATIVEWIKI. Cyan: best of B1–B5. Characters are split by their number of gold assessments (Appendix H).

	Input tokens	Pages	Input per page	Distinct per cutoff	Input per distinct
B5	33.13M	1,350	**24,541**	**6,970**	951
B1	69.23M	136	509,044	1,821	7,604
B2	**3.42M**	136	25,147	1,718	**398**
B3	5.08M	136	37,353	1,404	724

Table 25: Input tokens for all five cutoffs relative to what is published (B5: NARRATIVEWIKI; pages delivered over all cutoffs; distinct assertions as a mean per cutoff, and input per distinct assertion over their sum across the five cutoffs). Derived from Table 1 totals and the page counts of Table 23; not a separate run. Pages differ in length and kind across systems.

value superseded by a later one is shown as a closed interval rather than removed. Table 24 counts the transitions. B4, which reads NARRATIVEWIKI's graph but writes each page in one call, loses 12 and 14, within the range of the prefix-bounded systems, which locates the stability in rendering rather than in the graph.

What is published. At $t = 1..5$ NARRATIVEWIKI publishes 59, 117, 158, 191 and 236 character pages, 34, 71, 111, 147 and 196 relationship pages (one per pair of characters), one timeline page per volume and three shared pages for groups, places and objects. The comparison systems publish the character pages of the evaluated cast. Relations keep their history as intervals: at $t = 4$ the Anne–Gilbert block on Anne's page holds seven typed lines with paragraph citations (rival; romantically involved with; saved by; enemy, ended v1; comrade, since v2; friend, since v2; spouse, since v4), where long context prints one line, "Spouse of — Husband, former schoolmate and college rival". The cutoff-2 block lacks the spouse line, whose evidence is in book four (Appendix M).

Cost for the amount published. One NARRATIVEWIKI build serves every cutoff, while long context resends the whole prefix for every page (Table 25). Per distinct assertion NARRATIVEWIKI uses one eighth of long context's input. Prefix retrieval remains the cheapest: its 9.7-fold advantage

in total input tokens becomes 2.4-fold per distinct assertion and about equal per page.

Against LightRAG. On the 335 common fact checks, NARRATIVEWIKI alone conveys 94 and LightRAG alone 17 on the original text (both 175, neither 49); on the treated text 127 and 12. LightRAG remains ahead on precision, citation sufficiency and input tokens (Table 1).

Facts only NARRATIVEWIKI conveys. Gold facts that NARRATIVEWIKI conveys while B1 and B2 both miss them number 7 (15) on the original text (facts, with fact checks in parentheses): Anne's braids and changeable eye color, her mother's old letters, her longing for puffed sleeves, her story heroine Averil Lester, Gilbert rowing to her when her flat sinks, Rachel Lynde's "red as carrots", and Walter's hatred of Dan Reese. On the treated text they number 13 (20): six of the above and, among others, Anne realizing she loves Gilbert when he falls ill, Gilbert giving up the Avonlea school for her, and his refused proposal. Most are early-book details about main characters that a later, budget-limited page no longer has room for. The reverse, facts B1 and B2 both convey and NARRATIVEWIKI misses, numbers 13 (31) on the original text and 10 (15) on the treated text.

The other 40 characters. NARRATIVEWIKI trails on the other 40 characters. Much of that deficit lies in characters about whom it conveys no gold fact at any cutoff on the original text (Mr. Phillips, 0 of 10 fact checks; Charlotta, 0 of 4; Reverend Jo, 0 of 3; Roy Gardner, 0 of 3). Facts lost across cutoffs compare labels at adjacent cutoffs, so a labeling inconsistency between them also counts as a loss.

I A Formal Account of the Trade-off

The results of Section 6 share one cause: a claim, not its passage, is the generator's input. This appendix states the three consequences precisely. The statements are elementary; their use is to say which observed differences are structural, so that they will recur in any system built this way, and which depend on a particular build.

Notation. A stored claim is $c = (s, p, o, \mathrm{vol}(c), \mu(c))$: subject, predicate, object or value, the volume of its first evidence (`first_vol`) and a polarity $\mu(c) \in \{\textit{asserted}, \textit{presumed}\}$. The implemented schema stores no speaker and no finer modality (prediction, hypothesis, belief). For an entity e, let $I(e)$ be its verifier-accepted intervals and $I_t(e) = \{i \in I(e) : \mathrm{start}(i) \le t\}$ those eligible at cutoff t; the volume filter is $\phi_t(c) = \mathbf{1}[\mathrm{vol}(c) \le t]$. $G_t(e)$ denotes the gold facts about e eligible at t (Section 3).

Proposition 1 (stored claims are never evicted). For every entity e and cutoff $t < N$, $I_t(e) \subseteq I_{t+1}(e)$. Hence every structured line rendered for e at t is rendered at $t+1$, possibly with a closing volume added.

Proof. $\mathrm{start}(i) \le t$ implies $\mathrm{start}(i) \le t+1$. Structured fields are rendered deterministically from $I_t(e)$, and an interval that closes after t is shown as current at t and as a closed interval ("previously") once its end is reached, so it is never removed. □

Consequence. Under a fixed entity index, a gold fact conveyed by a structured line at t is conveyed at $t+1$. Losses can come only from prose regenerated per cutoff and from labelling differences between adjacent cutoffs; NARRATIVEWIKI loses 1 conveyed fact per text (Table 24). The guarantee is conditional on the entity index: with an index rebuilt per cutoff (Section 6.5) a page's entity can change between cutoffs.

Proposition 2 (a fixed page budget caps recall). Let a system write the page of e in one call under a budget of at most B sentences, each conveying at most k gold facts. Then at most kB facts are conveyed, so

$$\mathrm{recall}_t(e) \;\le\; \min\Big(1,\; \frac{kB}{|G_t(e)|}\Big).$$

Proof. Immediate from the count of conveyed facts. □

Consequence. For a character whose record keeps growing, the ceiling falls with t once $|G_t(e)| > kB$; Proposition 1 imposes no such ceiling on rendered claims. The shared template of B1–B4 allows at most $B = 22$ sentences (appearance 2–4, personality 2–6, history 3–12; Appendix E), while the eligible checks on the three central characters grow from 14 at $t = 1$ to 38 at $t = 5$ (Table 22). The bound is consistent with, but does not by itself explain, B1's fall on these characters (Figure 4a): k is not fixed, and the generator also chooses which facts to keep.

Proposition 3 (a volume filter is blind to modality). Let the renderer print (s, p, o) for every

claim with $\phi_t(c) = 1$, whatever $\mu(c)$. If some claim has $\mathrm{vol}(c) \leq t$, $\mu(c) \neq$ *asserted*, and (s, p, o) states a later gold fact $g \in F_t$, then the page at t discloses g.

Proof. $\phi_t(c) = 1$, so c is rendered, and the rendering states g. □

Consequence. Filtering by evidence volume is sufficient against disclosure only if every eligible non-asserted claim is either withheld from deterministic fields or rendered with its modality and speaker. On the treated text, a neighbour's prediction in book two was stored as a presumed marriage and printed as a spouse relation at $t = 2$ and $t = 3$ (4 of 220; Section 6.2). Prefix-bounded prompting keeps the passage, with speaker and mood, in the generator's input, and disclosed no assessed later fact.

Proposition 4 (routing by co-mention loses support). Let a scene record k with participants P_k produce a summary line a that is routed to every pair page $\{u, v\} \subseteq P_k$, and let a line be supported on a pair page only when u and v share an episode of the scene (the pair rule of Appendix B). If the episodes of the scene group the participants into disjoint sets $Q_1, \ldots, Q_m$, then of the $\binom{|P_k|}{2}$ pair pages that receive a, at most $\sum_j \binom{|Q_j|}{2}$ can support it.

Proof. A pair that shares an episode lies inside one Q_j. □

Consequence. A line can be true as a statement and unsupported on most of the pages it reaches, and the unsupported share grows with the participants per record and the episodes per chapter. In the reported build, scene records list a mean of 5.9 participants (first build: 9.0); on relationship pages 0.971 (99/102) of sampled lines are true as statements but 0.706 (72/102) are supported in page context (Table 15). Statement truth and page support should therefore be scored separately (Appendix J).

J Reporting Recommendations

The audits behind this paper suggest six practices for evaluating cutoff-bounded or spoiler-sensitive generation over published narratives. Each is followed by the result of this paper that motivates it.

K Validation of Automatic Judges

Claude Opus 5.5 finalized the reported recall, precision, citation-sufficiency and disclosure labels after reading the source text (Section 5). We evaluated two types of automatic component: the verifier used by NARRATIVEWIKI and potential screens for evaluation.

The verifier inside NARRATIVEWIKI. Negatives were produced by minimal word substitutions in source-verified facts, following Jeong et al. (2025), each confirmed false against the text and stratified by type (entity, attribute, negation, reversed relation). On these 81 items the verifier reached a balanced accuracy of 0.989 (1 of 46 unsupported admitted, 0 of 35 supported rejected). A second, harder set of 12 items pairs a quoted lie, a hypothetical, and a character's embedded story with true counterparts from the same passages. The verifier's balanced accuracy on this set is 0.750; it admitted every claim drawn from the embedded story. These measurements concern the reported verifier. Its logged `APPEARANCE` call contains a definition error (Appendix D); no corrected verifier was evaluated.

Screens considered for evaluation. On the same minimal-edit set, MiniCheck (Tang et al., 2024) reached a balanced accuracy of 0.718, rejecting nearly every negative but also more than half of the true facts, and a local instruction model, Qwen2.5-14B-Instruct (Qwen Team, 2024) (4-bit, served by Ollama), reached 0.867. On the hard set they reached 0.500 and 0.667. Consistent with the finding of Subbiah et al. (2024) that entailment classifiers transfer poorly to narrative, neither screen assigns evaluation labels. The same local model splits prose into atomic assertions, and string search retrieves candidate disclosure sentences. Claude finalized the sampled-atom and candidate labels against the source text.

L Disclosure Casebook

Table 27 illustrates disclosure channels observed in the reported builds and the first build of NARRATIVEWIKI, including lexical-screen hits that did not disclose later facts. Table 28 documents defects found and corrected during development. The examples reveal plot points of the evaluated series.

M Qualitative Example: One Page at Two Cutoffs

Table 29 compares Gilbert Blythe's page in the reported NARRATIVEWIKI site at cutoffs 2 and 4; volume 4 first establishes his marriage to Anne.

Recommendation	Practice	Why: evidence from this paper
R1 Pair every score with a paraphrase control	Report a closed-book floor and an aligned renamed-and-paraphrased condition beside any score on published fiction, with a recognition check of the treatment.	Renaming alone leaves the author identifiable in 0.625 of paragraphs; with paraphrase, closed-book recall falls from 0.683 to 0.003, and the treatment exposes X2's disclosures (8/88 to 17/88 unique later facts) and B1's citation decline (0.875 at t=1 to 0.300 at t=4).
R2 Add a single-call baseline on the same graph	Pair a multi-surface graph renderer with a baseline that writes pages in one call from the same graph and evidence.	B4, reading NARRATIVEWIKI's graph, reaches precision 0.965 and citation sufficiency 0.915 against NARRATIVEWIKI's 0.789 and 0.512: without it the gap would be attributed to extraction rather than to publication.
R3 Score statement truth and page support separately	Label sampled assertions both as standalone statements and in the page that prints them, and report both, by kind of page.	On relationship pages 0.971 (99/102) of NARRATIVEWIKI's sampled lines are true as statements but 0.706 (72/102) are supported on the page; page-context precision alone conflates misplacement with invention (Proposition 4).
R4 Audit construction scope against the prefix	Compare entities resolved over the whole series with indexes discovered from volumes 1..t at every cutoff before N.	Filtering claims leaves construction dependence intact: 0.390 of character pages at t=1 differ from a prefix-built index, 0.220 regrouped or absent (Table 17).
R5 Report retention across cutoffs, not only means	Count facts conveyed at t and lost at $t+1$, and stratify recall by how much the text says about a character.	Unweighted means (B1 0.848, NARRATIVEWIKI 0.802) hide that budgeted rewrites lose 4–24 conveyed facts across cutoffs while NARRATIVEWIKI loses 1 per text (Proposition 2).
R6 Validate the disclosure search and bound zero counts	Measure the candidate search on seeded paraphrases and pronoun references, and report character-clustered upper bounds with every zero.	The search retrieves 0.845 of named paraphrases but 0.309–0.355 when a pronoun on another character's page refers to the character (Table 14); a zero count at t=1 is compatible with a character-level rate up to 0.072 (0.221 at t=4).

Table 26: Six reporting recommendations for cutoff-bounded generation over published narratives, each with the result that motivates it.

The reader picks a cutoff on the home page before seeing any content, and each cutoff's pages use only claims whose evidence volume is at most that cutoff (Figure 1a shows the stored intervals behind the relationship block). The pages themselves were not annotated; precision labels in Section 6 assess sampled atoms of the underlying pages. The upper block of the table shows the intended behaviour of intervals; the lower block shows four defects of the reported build.

N Contamination Diagnostics and the Treated Text

A generator may recognize published novels (Chang et al., 2023). We compare aligned original, names-only and renamed-and-paraphrased text, assessing recognition separately from later-fact disclosure and cutoff compliance. Renaming follows prior practice (Gupta, 2025; Zhang et al., 2024); cross-condition differences motivate contamination diagnostics (Dekoninck et al., 2024). Because a treatment can damage the narrative (Sun et al., 2025), we examine gold-fact fidelity and document errors. The procedure aims to preserve facts, attribution, reveal order and cross-volume coreference, but has no independent validation of complete fidelity. Reduced recognition does not rule out memorization (Cheng et al., 2025).

Treatment. The treatment operates paragraph by paragraph, retaining identifiers, chapter boundaries and scene breaks so citations remain aligned.

Step 1, names. A reviewed map of 239 entries replaces character, family and local place names with invented forms. Longer forms precede shorter ones in whole-word replacement, with consistent full names, surnames and nicknames. Mechanical checks require stand-ins to be absent from the source names and every part of a mapped full name to be mapped. An initial map failed both checks after one volume: a main character's stand-in matched two minor characters' first names, and a mapped full name left its 148 bare-first-name occurrences unchanged. We rebuilt the map and discarded that treated volume. Recurring minor characters and local places are renamed; countries and distant cities remain. A surname also used as a

System	t	Surface	Delivered text	Source and reveal volume	Verdict
X2	2	character page (Priscilla Grant)	"She entered Redmond College in Kingsport, taking up the Second Year work alongside Anne."	Anne studies at Redmond: `v03:c01:p0013`, reveal 3	stated; the later text was in the input with an instruction not to use it
X2	2	character page (Jane Andrews)	"Later, Jane accepted a teaching position in Winnipeg, West Canada, where she became engaged to a wealthy English millionaire."	Jane leaves for the West: `v03:c14:p0030`, reveal 3	stated
X2	4	character page (Susan Baker)	"Role: Housekeeper at Ingleside"	Anne lives at Ingleside: `v05:c01:p0001`, reveal 5	stated
X1	1	character page (Billy Andrews)	"Too bashful to propose to her himself, he sends his sister Jane to propose marriage to Anne on his behalf."	`v03:c08:p0012`, reveal 3	stated; no text in the input, so the source is the generator's memory
X1	4	character page (Little Jem)	"When Gilbert's medical practice expands, the Blythe family moves to Ingleside in Glen St. Mary."	`v05:c01:p0001`, reveal 5	stated (memory)
NARRATIVEWIKI	3	relation summary on a character page (Phil Gordon)	"Phil later serves as a bridesmaid for Anne."	Anne marries Gilbert: `v04:c04:p0000`, reveal 4	implied: a reader infers Anne's wedding, and by the end of book three she is engaged to Gilbert (`v03:c41:p0036`). Written from eligible text and wrong: in `v03:c39:p0039` Anne is Phil's bridesmaid
NARRATIVEWIKI	1, 2	relation line on two character pages and a pair page	"Anne Shirley — Romantically involved with — Gilbert Blythe — harbors romantic feelings for her · since v1", citing `v01:c25:p0055`	Anne marries Gilbert: `v04:c04:p0000`, reveal 4	uncertain: in the cited passage a friend reports that Gilbert kept a rose Anne dropped; the label reads towards the later romance
B1	2	character page (Rachel Lynde)	"…allowing Anne to attend Redmond College"	decision in book two, attendance `v03:c01:p0013`	uncertain
B4†	2	quote on a character page (Ruby Gillis)	"Oh, she does look really dead." (`v01:c28:p0013`, said of Anne playing a corpse)	Ruby Gillis dies: `v03:c14:p0033`, reveal 3	not a disclosure; an entailment screen scored it 0.52
B1–B4	3	title of a character page	"Reverend Jo", the name under which the evaluated cast lists the character	book three has "the Rev. Jo" (`v03:c39:p0039`); spelled out first in book four	expanded name form supplied by the evaluation's page list; no fact revealed
NARRATIVEWIKI †	2	existence and title of a character page (`walter`)	a page "Walter" describing Anne's father, wrongly	the entity is Walter Blythe of book five (127 of its 130 mentions)	later-anchored page; no fact revealed; absent from the prefix build and corrected in the reported build
NARRATIVEWIKI	2	existence of a character page (`susan`)	an empty page titled "Susan"	the entity is Susan Baker of books four and five (3 of its 204 mentions by book two)	later-anchored page; no fact revealed
NARRATIVEWIKI	1	codex page (groups)	"Conservatives"	book one has the singular	plural form; lexical screen hit, no fact revealed

Table 27: Cases from the reported builds and the first build of NARRATIVEWIKI, by disclosure channel. Each row gives the delivered text, the source and reveal volume of the fact concerned, and the verdict. †: the first build of NARRATIVEWIKI, or B4 on its graph.

System	t	Surface	Delivered text	Cause or correction	Verdict
baselines	1	list of pages and their titles	pages for eleven characters whose names do not occur in book one	the page-selection rule matched any single word of a name	disclosure by page title; rule changed to "a gold fact by t or the full name in $D_{\leq t}$"
NARRATIVEWIKI	3	page title and address	a page for Leslie Moore, who first appears in book four	a book-three mention "Will Leslie" was resolved to her	disclosure by title; resolution fixed
NARRATIVEWIKI	1	relation qualifier	a qualifier written from book-two evidence on a relation opened in book one	qualifier now taken from evidence no later than the interval's first volume	premature detail; fixed
NARRATIVEWIKI	all	shared navigation	names of later characters in a menu shared by every cutoff	the menu listed all pages of the site	disclosure by menu; shared navigation now names no character

Table 28: Development defects and the corrections applied before generating the reported evaluation cells.

compass direction is replaced only in person-name contexts.

Step 2, paraphrase. The generator rewrites each renamed paragraph under instructions to preserve facts, events, numbers, places, ordering, names and dialogue speakers, with no additions or summarization. Reapplying the map replaces any restored source names. Nonempty rewrites between half and twice the source length are accepted; otherwise the renamed paragraph is retained and counted as unparaphrased. Residual source names are counted separately. In an initial treated volume, approximately one paragraph in 25, containing one word in 12, retained at least four fifths of its four-word sequences. We therefore request one retry when at least half the source's four-word sequences remain. A second near-copy is retained but counted as verbatim. Failed model calls stop the run, preventing silent fallback.

Step 3, gold. The same map transforms the original gold facts and cast; facts are neither added nor removed, and reveal volumes are unchanged. All systems in Section 5 are rebuilt with unchanged code, configuration and prompts. A names-only variant omits Step 2 and is used for recognition probes, rather than complete system rebuilding.

What the condition can and cannot show. If systems mainly use supplied evidence, much of their original-text coverage may survive the treatment. Reduced recognition may instead reduce closed-book coverage, where no source text is supplied. These are diagnostic expectations, not an identification strategy for each system's memory contribution: rewriting also changes wording, entity resolution and sometimes meaning.

A prefix-bounded system can disclose later facts through extraction or inference errors, generation or residual recognition. A treated-text disclosure by the full-text diagnostic still demonstrates a failure to obey the cutoff while later evidence is available. A model can recognize a plot without its original names, so the treatment cannot establish the absence of memorization.

The treated text. The treated series contains 8,489 paragraphs and 428,122 and 467,246 words before and after treatment. Of these paragraphs, 8,468 (99.8%) were paraphrased, with the remainder renamed only; 1 remained near-copies after retrying. Paraphrase lengthens the series overall, although each accepted rewrite stays within the length bounds of Step 2. Whole-name checks find no mapped source name unchanged. A broader search finds 27 inflected or derived source-name occurrences in 26 paragraphs: chiefly family-name plurals ("the Gardners"), plus a college-derived adjective and a sentence explaining a name's origin. Recognition probes include these residual forms.

Some residuals reach generated pages. Across 1,597 character pages from seven systems, the name explained in the source sentence appears in 25 pages of six systems. A family name retained in a plural and a chapter title ("The Gardners' Call") appears on 9 pages of four systems. Three other family names and the college name, each from a residual form, appear on seven pages. A further closed-book match is a common surname in an invented story.

Another source name has no literal residual: Six pages from prefix retrieval and the full-text diagnostic print "Prince Edward Island", absent from the treated paragraphs and chapter titles. The renamed island's neighboring provinces remain, permitting geographic inference. This output therefore does not distinguish inference from residual recognition.

Field	Cutoff 2 (*Anne of Avonlea*)	Cutoff 4 (*Anne's House of Dreams*)	Mechanism
Intended behaviour			
Role	teacher [v1 ch. XXXVI P24, v2 ch. I P55, +7]	country doctor (since v4) [v4 ch. 14 P8]; *previously:* teacher (v1–v2); Daily News employee (v3)	Supersession: an earlier value becomes a closed interval when a later one opens.
Titles	president (since v2) [v2 ch. VI P8]	adds Captain of the Freshman football team (v3), president of the Freshman Class (v3), doctor (v4)	Accumulation: volume-dated titles are appended; none is evicted.
Family	(no line)	Anne Shirley (spouse)	Withholding: no family claim has evidence by volume 2.
Anne–Gilbert block	rival; romantically involved with; saved; enemy (ended v1); comrade (since v2); friend (since v2)	the same six lines, with more citations, and *Spouse of* — husband and wife (since v4) [v4 ch. 1 P5, ch. 2 P7, ch. 3 P29, +26]	History is kept as intervals; long context prints one line ("Spouse of — husband, former schoolmate and college rival").
Block prose	ends with the two as close friends	ends with the marriage	Each cutoff's prose is generated from its own filtered claims.
Defects of the reported build			
(i) Relation label	*Romantically involved with* — harbors romantic feelings for her, open from v1 [v1 ch. XXV P55]	unchanged	The label is stronger than its qualifier and evidence: NARRATIVEWIKI's uncertain disclosure verdicts (Section 6.2).
(ii) Line keys in prose	none	prose prints "[R3, R4, S1, S4]" and "[R2, R7, S41, S67]"	The renderer resolves keys into the *Sources* line but leaves them in the prose: 5 of 341 non-timeline pages at t=4.
(iii) Age	nearly fourteen [v1 ch. XV P24]	nearly fourteen, beside the marriage	An attribute with no later extracted value stays open; elapsed years are not inferred.
(iv) Second entity	—	a block "Shirley" holds *Rival of* [v1 ch. XVII P36]	The build merged two of three Anne entities; the bare surname remains separate (Appendix G).

Table 29: Gilbert Blythe's page in the reported NARRATIVEWIKI site at cutoffs 2 and 4, transcribed from the delivered pages (screenshots are released with the code). Citations are printed as the site prints them (volume, chapter, paragraph; "+n": further citations). Cyan: intended interval behaviour; magenta: defects.

Fidelity. An unblinded LLM annotator checked all 111 gold facts against their treated evidence paragraphs. It judged all 111 still expressed, one with altered wording: a sleeve style is renamed in its evidence paragraph but given its source name in the next paragraph. This asks whether the treated passage retains the fact. The stricter paragraph-alone audit from Appendix B, yielding 85, 23 and 3 for the original text, was not repeated.

Errors also occur outside gold evidence. One paraphrase changes two children's ages into an interval of minutes, after which prefix retrieval reports twins born six minutes apart. The statement counts as supported against the supplied treated text. Treatment errors found during analysis are documented here and left in place.

Recognition. On 200 aligned paragraphs spanning five volumes, the generator restores a masked character name with accuracy 0.165 on original text and 0.000 on treated text. Title identification is 0.820, 0.325 and 0.110 for original, names-only and combined treatment, respectively; author identification is 0.865, 0.625 and 0.195. Renaming reduces title recognition more than author recognition. Adding paraphrase reduces both further, to approximately one third of their names-only levels, but roughly one treated paragraph in nine is still identified as belonging to the series. Only recognition was probed for names-only text; complete systems were rebuilt on original and combined-treatment text.

Systems on the treated text. Table 1 compares original and treated systems; Table 10 reports every cutoff. Gold facts, reveal volumes and labeling rules are transferred from the original; inverse name mapping allows the same per-fact tests. Claude Opus 5.5 finalized the AI labels for all systems on both texts, including precision and citation sufficiency.

Each condition has one build. Differences in

Text condition	Title	Author
Original	0.820	0.865
Names remapped	0.325	0.625
Remapped and paraphrased	0.110	0.195

Table 30: Recognition ablation on 200 aligned paragraphs. Values are the share for which the frozen generator identifies an Anne title or its author. Renaming and paraphrasing both reduce recognition, while residual recognition remains. Cyan highlights the combined treatment. This probe does not establish that a wiki fact was memorized.

entity resolution and paraphrase fidelity preclude interpreting small cross-condition changes as isolated memory effects. The closed-book diagnostic has a much larger recall reduction, from 0.683 to 0.003, with 1 of 335 matching fact–cutoff assessments remaining: a death also present in its invented story. This is consistent with recognition contributing substantially to original-text closed-book recall.

What the closed-book generator writes. Given stand-in names without text, the generator still writes full pages. Of 200 sampled assertions, 162 are unsupported. The stand-in for a shy book-one farmer becomes a sea captain founding a cult; a heroine's friend becomes a duelist at a magical academy; later cutoffs introduce a street gang and a starship. Supported judgments concern generic statements about gender, status, siblings or traits. No assertion carries a citation.

The disclosure by NARRATIVEWIKI. At $t = 2$ and $t = 3$, NARRATIVEWIKI lists both leads as spouses, a book-four fact (4 of 220, counting the marriage for both characters at both cutoffs). These are two reciprocal subject facts from one prediction-to-marriage failure mechanism, rather than four independent errors (Table 20). Another 4 verdicts are uncertain, involving the same courtship line as in the original. The marriage citations point to a book-two neighbor's prediction. Extraction stored a presumed marriage at volume two, and rendering turned presumed spouses into spouses. B4, using the same graph, prints "anticipated future marriage" at $t = 2$, labeled implied disclosure (2 of 220).

The original build did not extract this relation from the corresponding passage. The treated passage is cutoff-eligible, but the claim loses the distinction between a character's prediction and an established event. Volume filtering cannot repair that attribution and modality error.

Long-context prompting on the treated text. Mean recall is 0.845 on treated text versus 0.848 on original text; precision is 0.920 versus 0.929. Both treated estimates are lowest at $t = 4$ (Table 10). The close means do not establish equivalence from single builds.

Citation sufficiency is 0.875 and 0.675 for one and two volumes, then falls to 0.425, 0.300 and 0.375, versus 0.700, 0.590 and 0.550 on the original. Most insufficient citations accompany supported assertions. Many retain the paragraph number but lose the volume or chapter, pointing, for example, from a volume-one fact to the corresponding identifier in volume three. Such identifiers pass the harness's existence check.

The cause is untested. Familiarity with the original might help locate evidence, or paraphrased passages might be harder to distinguish. Neither explanation is isolated by this comparison.

The full-text diagnostic on the treated text. Despite the cutoff instruction, X2 discloses 23 of 220 assessed later facts on treated text, versus 8 of 220 on the original. Per-cutoff counts at $t = 1..4$ are 8/88, 8/65, 6/45 and 1/22; original counts are 0/88, 6/65, 0/45 and 2/22. At $t = 1$, the guardian's page describes the neighbor's widowhood and move, the heroine's college attendance and her marriage "years later". It cites volumes two to four, whose identifiers the harness accepts because all volumes are supplied. Reduced recognition therefore does not make the instruction reliable; observed disclosure is higher at the first cutoff. Familiarity with the original's book boundaries is one untested explanation.

Treated recall is lower at every cutoff (0.744 versus 0.814); mean precision is also lower (0.845 versus 0.910). Most unsupported statements are misattributed biographies, and citation errors also omit volumes. The heroine's $t = 1$ page returned malformed JSON three times, closing a list with the wrong bracket. We corrected that bracket in the recorded first response without another model call (Appendix P).

Namesakes made by the name map. Whole-word renaming creates two collisions observed in the systems receiving the most text, B1 and X2. First, one stand-in replaces a shared first name across characters, making a book-one minor character a namesake of a schoolmate; passages about

one are then cited for the other. Second, a book-two twin's stand-in is a short form of the stand-in for the heroine's romantic name, also borne by her friend's daughter in book four. With book four in context, the twin's page describes the friend (B1 at $t = 4$; X2 at $t = 2$, 4 and 5) or the heroine (X2 at $t = 3$), whereas original-text builds describe the twin.

This page contributes unsupported assertions and two stated X2 disclosures at $t = 3$. Neither mechanical map check detects these collisions. They are treatment defects retained in the reported system scores.

O Development Observations

Choice of generator. We sent stored claim-extraction prompts from a commercial development series to three Gemini Flash models, then checked the outputs of two of them, 3.1 Flash-Lite and 3.6 Flash, against their cited paragraphs. No numerical results from that series are reported (Appendix C). Qualitatively, Gemini 3.1 Flash-Lite returned more facts than Gemini 3.6 Flash, with a lower share fully supported and some unsupported claims; no unsupported claims were observed in the reviewed 3.6 Flash outputs. We selected 3.6 Flash for all stages and comparison systems. Model choice also affects entity discovery: from the same volume-one Anne candidate list, Flash-Lite retained 83 (50 characters) entities and 3.6 Flash retained 92 (54 characters). This illustrates an effect on the entity inventory before page generation.

Stories within the story. Characters tell stories, invent companions and make false confessions. Entity discovery admitted figures from embedded stories, including a story-club heroine and a child's imagined cave lady, and verification accepted claims about them (Appendix K). Local evidence can therefore be insufficient to distinguish the main narrative from an embedded or imagined account.

Reveal volumes need an earliest-statement check. Disclosure assessment found 2 of 90 draft facts assigned reveal volumes later than their earliest supporting statements. Pages containing these facts were consequently eligible. Reveal-volume annotation requires searching for the earliest support, rather than retaining the first passage encountered by the annotator.

Implicit temporal progression. Gilbert's age, *nearly fourteen*, is extracted from volume one and remains on his cutoff-4 page alongside the later marriage (Table 29). No extracted age supersedes it, so its interval stays open. The graph records explicit changes; it does not advance ages with narrative time. Reveal volumes order evidence rather than elapsed story years. A historical age should therefore be marked as an observation at a particular story time unless a supported update is available.

Cross-claim inference. Some updates require combining claims rather than extracting one explicit statement. As an illustrative example, suppose Anne is two years younger than Gilbert and marries him at age 25. The premises imply that Gilbert is 27 at their marriage. These ages are hypothetical, not verified annotations of the displayed example. The current pipeline does not systematically derive such updates. An extension would need to retain the supporting premises, align their narrative times and expose a derived claim only after all required premises become available at the reader's cutoff.

Coreference across extraction windows. Pronouns and other referring expressions can link a character or fact across sentences and scenes without repeating a name. Extraction may miss these links or attribute a claim to the wrong person when the relevant context lies outside its window. Both backward and forward references require discourse context, while a later identity revelation must remain unavailable at an earlier cutoff. These are qualitative failure modes; no error rate for them is measured here.

P Reproducibility and Release

Freeze. Before each block of runs, the source code, configuration, and scripts are hashed into a freeze record that every runner checks before making a call. Any change requires a new record and an account of which results it invalidates. Changes affecting a reported number are logged with the previous and revised values and the cells rebuilt or relabeled, including the two label audits in Section 5. Sampling seeds are fixed (assertion samples: seed 0).

Provenance. A shared client caches every model response under a hash of its provider, model, prompt, and parameters; each run writes a manifest

and call log. Every number in the paper maps to a measurement-ledger row identifying the producing command and run. All generative calls of the systems and of the treatment used Gemini 3.6 Flash (Vertex AI, temperature 0.4, `thinking_budget` 0), between 28 September and 7 October 2026. The provider's model name exposes no finer revision. Cached responses reproduce the released artifacts exactly; fresh calls need not produce identical generations. On the treated text, one page of the full-text diagnostic returned a list with an incorrect closing bracket in three attempts. The runner corrected the bracket in the recorded first response and applied its normal validation and rendering; no page was regenerated. Appendix Q lists every generative stage with its logged call and prints two exchanges in full; the release holds one complete logged exchange per stage, with system prompt, input and output. Claude Opus 5.5 finalized all reported AI annotation labels after evidence review. The prompt and response defects documented in Appendix Q remain part of the archived run configuration. The reported results are not measurements of a repaired prompt builder or citation validator.

Commands. With the corpus parsed, the reported artifacts are produced by:

- the reported NARRATIVEWIKI build of all five cutoffs:
  `run_series.py anne@v2 --upto 5` (the first build is the series `anne`);
- a comparison cell at cutoff t:
  `run_baseline.py anne {X1|B1|B2|B4|X2} --upto t`, with `--chunk-words 150 --budget-tokens 20000` for B2 and `--graph anne@v2 --budget-tokens 20000` for B4, and `b3_lightrag.py anne --upto t --mode mix`;
- the assertion inventory and sample of a cell:
  `assertion_inventory.py S --upto t` and `assertion_precision.py S --upto t --n 40 --population all`;
- disclosure candidates:
  `m4_tree.py S --upto t`;
- the lexical screen:
  `artifact_exposure.py S`;
- the pair rule and the timeline citations applied to a cell's labels:
  `apply_pair_rule.py S B5`;
- table values:
  `paper_numbers.py`; their spread over cutoffs: `summary_stats.py`;
- cost of the cells on disk:
  `run_cost.py S --final`.

Release. An anonymized archive provides the submission materials; the repository address is https://anonymous.4open.science/r/NarrativeWiki-Anon-341E/. It contains the code, configuration, dependency specification and freeze records; prompt templates and logged exchange excerpts; gold facts with evidence identifiers and reveal volumes; and AI evaluation label files (recall, annotated precision samples, citation sufficiency, disclosure verdicts with the sentences assessed, and audit corrections). It also includes per-run cost summaries, every system's generated wikis at every cutoff, both builds of NARRATIVEWIKI, and the human-review package (554 items with source context: every gold fact and samples of the recall, precision and citation labels), plus the two reviewers' label files, accompanying records, provenance manifest, aligned decision audit and recomputed agreement. The reviewer labels are entirely human-produced. The source texts are public-domain and are fetched from Project Gutenberg by identifier.

Q Prompts and Model Exchanges

Every generative stage calls the generator through one client that records the system prompt, the user prompt and the raw response. Table 31 lists the stages whose first successful call from the run behind the reported results is exhibited in the release; all used Gemini 3.6 Flash (Vertex AI, temperature 0.4, `thinking_budget` 0). This appendix prints two of those exchanges in full, the two a reader needs to reproduce the method (same-volume arbitration and the paraphrase that makes the treated text), and the four defects that the logged prompts and responses carry. The complete prompts and every call log are released (Appendix P); the excerpts below are verbatim, with elisions marked. Evaluation labels were not produced by these stages (Appendix B).

Same-volume arbitration

Two values of one attribute from one volume are treated as an extraction conflict (Section 4). A `flag` resolution keeps the higher-confidence value provisionally (here 11), so the interval sequence has no gap, and records the conflict.

Stage	Role	Tokens in / out	Run
NARRATIVEWIKI *build*			
Entity typing	Types each mined name candidate (gazetteer)	518 / 39	`20261004T193719-gazetteer-v1-5`
Alias adjudication	Decides whether two surface forms are one entity (gazetteer)	548 / 50	`20261004T193719-gazetteer-v1-5`
Claim extraction	Extracts dated, quoted claims about one character from one evidence window	8,884 / 734	`20261005T070204-extract-v5-5`
Scene extraction	Records scenes, participants and events over a chapter span	5,729 / 202	`20261004T195945-scenes-v5-5`
Infobox pass	Long-context pass for gender, age and origin	93,403 / 490	`20261005T070216-infobox-v1-5`
Backstory pass	Long-context pass for background facts	135,267 / 309	`20261005T070229-backstory-v1-5`
Arbitration	Resolves a same-volume conflict between two values	258 / 55	`20261005T070243-graph_build-v1-5`
Verification	Checks a stored fact against its own cited evidence	2,833 / 5	`20261005T053635-verify-upto5`
NARRATIVEWIKI *publication*			
Page prose (appearance)	Writes a physical description from the cutoff-filtered claims	1,045 / 174	`20261005T070347-synthesize-upto5`
Codex summary	Writes the one-sentence summary of a shared codex entry	1,974 / 69	`20261005T070527-site_build-upto5`
Comparison systems and diagnostics			
B1 page	Long context: one character page from the whole prefix	559,824 / 5,688	`20261001T043209-baseline-b1-v4`
B2 page	Prefix retrieval: one character page from retrieved passages	24,083 / 2,477	`20261002T095544-baseline-b2-v1`
B3 index	LightRAG's own entity and relation extraction while building its index	3,243 / 1,022	`20260929T225346-baseline-b3-v5`
B3 page	LightRAG: one character page from LightRAG context	17,438 / 2,191	`20261001T142328-baseline-b3-v5`
X1 page	Closed book: title and character name only	451 / 1,087	`20260930T214444-baseline-x1-v5`
X2 page	Full text: all five volumes and the cutoff instruction	689,967 / 2,449	`20260930T222548-baseline-x2-v1`
Treated text and recognition			
Paraphrase	Paraphrases one renamed paragraph	385 / 209	`20261003T165238-decontaminate-v5`
Memorization probe	Name cloze or title identification on one paragraph	227 / 9	`20261003T165318-probe-memorization-v1-5`

Table 31: Every exhibited model stage: the first successful call of the run behind the reported results. Runs are under `anne@v2/` (build and pages) or `anne-decon/` (treatment and probe) in the release. B4's page prompt, the other prose sections and the shared-page summary check are in the release only.

User prompt (system prompt in the release)

```
Character: anne-shirley
Attribute: AGE
Volume: 1

Fact A: value="11", confidence=1.00, evidence: "I was
  eleven last March," said Anne"
Fact B: value="13", confidence=0.99, evidence: "It's such
  a solemn thing to be almost fourteen, Marilla."; ""Just
  think, Diana, I'm thirteen years old today," remarked
  Anne in an awed voice."

Which fact is correct? resolution must be one of keep_a,
  keep_b, keep_both, flag, plus a one-sentence rationale
  citing the evidence.
```

Model output

```
{"resolution": "flag",
 "rationale": "Fact A states Anne was eleven based on an
  earlier passage, while Fact B shows her turning thirteen
  later in the same volume, indicating a progression of
  time rather than a single static age."}
```

Paraphrase for the treated text

Each renamed paragraph is paraphrased in one call, keeping its identifier.

System and user prompt (paragraph abridged)

```
SYSTEM: You rewrite passages of fiction in new words for a
   research corpus. You never add, remove or explain
  anything.

Rewrite the paragraph below in different words and
  different sentence structure.
Rules:
- Keep every fact, event, number, date and place, and keep
   them in the same order.
- Keep every personal and ship name exactly as written. Do
   not introduce any name that is not in the paragraph.
```

```
- Dialogue stays dialogue, spoken by the same person. Use
  the same quotation marks as the original (" " stay " ").
   Only spoken words go inside them; a speech tag such as
  "said I" stays outside.
- Keep the narrator's person (first person stays first
  person).
- Do not summarise, shorten substantially, or add
  commentary.
Return only the rewritten paragraph, nothing else.

Paragraph:
It was a clear, apple-green evening in May, and Seven
  Gulls Harbour was mirroring back the clouds of the
  golden west between its softly dark shores. [...]
```

Model output (abridged)

```
The May evening was transparent and apple-green, as Seven
  Gulls Harbour reflected the clouds from the golden west
  amidst its gently darkened banks. [...]
```

Defects carried by the logged prompts and responses

The four defects below are part of the reported runs. They are documented, not repaired; Section 7 and Appendix I relate each to the results.

D1 Implementation notes used as field definitions (extraction and verification)

The extraction prompt defines STATUS and APPEARANCE with internal revision labels and code paths, and the verifier is told to judge an APPEARANCE fact by that same text:

Verification user prompt (excerpt)

```
Fact: APPEARANCE = "little pale face with seven
  freckles on the nose" -- evidence: [v02:c27:p0041]
  [...]
Definition of APPEARANCE: subsume (Phase 22 A2): drop
   a value that is redundant next to a longer sibling
   ('hair' next to 'long brown hair') or a bare
  single-word noun standing alone -- see graph/
  canonicalize.py::subsume_values. Deterministic, no
  embed call.
```

The exhibited claim is accepted; the example alone does not measure the defect's effect.

D2 Responses depart from prompt rules the code does not enforce (infobox pass)

The volume-1 infobox response gives a GENDER quote that does not contain Anne's name, and two AGE values (11 and 13). The code applies the evidence gates of Section 4, not these two prompt rules; the two ages go to arbitration, shown above.

D3 Early-volume prompts carry later aliases

Alias lists come from the whole-series entity index, so the volume-1 infobox call already names Anne by her married names:

Infobox user prompt, volume 1 (excerpt)

```
Character: "Anne Shirley" (aliases: Anne, Anne-girl,
  Mistress Blythe, Mrs. Blythe, Mrs. Doctor, Mrs.
  Doctor, dear, Mrs. Dr. dear, Queen Anne, the Anne-
  girl, young Mrs. Doctor)
Volume 1.
```

This is construction dependence reaching the generator's input (Section 6.5); the comparisons of this paper do not isolate it.

D4 Residue of one prompt serving every corpus

The typing prompt lists a type unused here (TECH) and, as logged, repeats one example sentence three times; the extraction prompt's JSON example uses a character from another series ("Shin"); the paraphrase rule keeps "personal and ship" names, which has no effect on this series.